



Study of primary instability of thick liquid films under strong gas shear

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ABSTRACT

The initial stage of downward co-current annular flow was studied using two optical techniques: brightness-based laser-induced fluorescence technique and shadow technique. The experiments were conducted in a circular pipe and in a rectangular duct. The working area was represented by the first few centimetres from liquid inlet; the latter was organised as a tangential slot. The process of formation of disturbance waves was found to consist of three stages: formation of regular initial two-dimensional high-frequency waves; fragmentation into localised irregular three-dimensional waves and formation of large-scale quasi-two-dimensional disturbance waves. All the stages occur closer to the inlet at higher gas velocities and lower liquid flow rates. It was found that the initial two-dimensional waves appear at the initial area of thick film in the vicinity of the inlet. Spatiotemporal measurements were conducted and spectral and statistical approaches were applied to study the characteristics of the initial waves at the linear growth stage. The obtained characteristics were compared to the linear stability calculations; the comparison showed satisfactory agreement for the frequencies and velocities of the initial waves after taking into account non-equilibrium film thickness at the inlet.

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1. Introduction

Annular-dispersed flow is commonly referred to as the joint flow of a liquid film along the inner walls of the channel and a high-velocity gas stream in presence of liquid droplets entrained from the film surface. This flow regime occurs in a variety of industrial facilities and power engineering, from compact heat exchangers to large-diameter pipes for the production and transportation of hydrocarbons. For the practical purposes, the integral characteristics of such a flow (average film thickness, entrained fraction, pressure drop, heat and mass transfer) are usually predicted by empirical models. As a rule, such models work properly only in the range of conditions in which they were created; their extrapolation onto a broader range of conditions yields large scatter between the measured and predicted values. A solution of the problem - and a distant-perspective aim of our research - is to develop the physically-based models of annular flow.

Creation of such models is hampered by the complexity of the flow, manifested in coexistence of numerous hydrodynamics phenomena of various scales. In particular, the liquid film surface

is dominated by fast large-scale disturbance waves (see the detailed description of these structures in Hewitt & Hall Taylor 1970, Hewitt et al. 1990, Azzopardi 1997, Alekseenko et al. 2014, Cherdantsev et al. 2014). Presence of these structures is necessary for the liquid entrainment; they carry the major fraction of liquid and strongly affect the integral characteristics of the flow. Thus, creation of new physically-based models, ensuring effective and safe operation of industrial devices, has to start with prediction of the dynamics and properties of the waves on film surface. One of the widely used ways to do so is to derive the empirical correlations relating the measured properties of the disturbance waves (velocity, frequency, amplitude, intermittency etc.) to the flow parameters. Many of such correlations are summarised in a recent review paper by Berna et al. (2014). The empirical correlations predicting the disturbance waves properties possess the same flaws as those predicting the integral characteristics of the flow. In our opinion, the physically-based models of the integral characteristics of the flow should be based on physically predicted properties of the disturbance waves.

Another approach to predict the disturbance waves properties is related to purely numerical models, appearing in recent times (Shine et al. 2013; Höhne & Hansch 2015; Kumar et al. 2016; Sato & Niceno 2017; Yang et al. 2017). Such modelling requires im-

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mense computational power due to large Reynolds numbers of both phases, unknown complicated shape of the interface with high density ratio and necessity to resolve extremely small spatial and temporal scales in a large three-dimensional spatial domain. To reduce the calculation domain, periodic boundary conditions are often used, which leads to neglecting the development of the waves. The modelled shape and characteristics of the disturbance waves are usually different from what is observed in the experiments with high enough spatial resolution (see, e.g., Cherdantsev et al. 2014). An exception among these works is the recent paper by Fan et al. (2018), in which the whole inner volume of the pipe serves as the computational domain, starting from the tangential-slot inlet. This model qualitatively reproduces the main stages of waves development, observed in the experiments (Alekseenko et al. 2015). At present, only one combination of flow parameters was modelled by Fan et al. (2018) due to strong demands in the computing resources.

The physically-based theories of waves in annular flow do not require as much computation power, but require simplifying basic assumptions, which should be validated experimentally. With rare exceptions (such as the model by Hall Taylor et al. 2014), existing theoretical studies of disturbance waves are limited to neutral stability analysis. Such analysis (see chapter 6.5 in Hewitt and Hall Taylor, 1970) yields the wavelength of disturbance waves of order of 0.1 mm (see Fig. 6a in Liu & Bai 2017), which makes a very strong (3 orders of magnitude) underestimation: a typical distance between the disturbance waves in the experiments is about 100 mm (see, e.g., Fig. 16 in Dasgupta et al. 2017). Stability analysis was modified for long-wave perturbations which are assumed to develop into the disturbance waves by, e.g., Hanratty & Hersman (1961), Andreussi et al. (1985). Due to significant empiricism, the calculated conditions of neutral stability are in good agreement with the regime maps of the existence of disturbance waves in a horizontal flow. However, the model does not provide any information on further waves evolution and, in its present state, cannot be used to predict the waves properties.

Alekseenko et al. (2015) used Brightness-Based Laser-Induced Fluorescence (BBLIF) technique to study spatiotemporal evolution of liquid film next to the inlet. Their observations show that in process of disturbance waves formation the spatial scale of the waves undergoes a cascade of changes. In the vicinity of the inlet, where the film surface is smooth and stability analysis would be appropriate, high-frequency initial waves appear (see also Zhao et al. 2013). Their wavelength is two orders of magnitude shorter than that of disturbance waves, but one order of magnitude larger than that predicted by simplified inviscid model (Hewitt and Hall Taylor, 1970). If the gas and liquid flow rates are high enough, these initial waves undergo multiple irreversible coalescence events promptly after their appearance. As a result of the coalescence, structures of much larger longitudinal size with strongly perturbed surface - i.e., disturbance waves, - appear at the first few centimetres from the inlet. Further downstream the disturbance waves coalesce with each other, which reduces their frequency (Hall Taylor and Nedderman, 1968) and, hence, further increases their "wavelength". Very far from the inlet (at the distances of over 150 pipe diameters, according to Wolf et al. 2001), the coalescence becomes a rare event and the disturbance waves travel with constant speed over long distances (Hall Taylor et al. 1963).

It is obvious from the above description that the results of stability analysis should not be compared to the distance between the disturbance waves. Nonetheless, the stability analysis is necessary to model the initial stage of film evolution, namely, the aforementioned initial waves. The model should not be limited by obviously rough approximations such as the inviscid model. Such a model might make a decent first step towards the physical modelling of the disturbance waves. To validate such analysis, the quantitative

experimental study of the initial waves is required. The characteristics of the initial waves (such as wave number, phase velocity and growth rate) should be measured at the linear growth stage of their evolution and compared to the properties of the waves of maximum growth predicted by the theory. A comparison of such sort was previously conducted by Alekseenko et al. (2009), but for much lower gas and liquid flow rates. The data obtained by Alekseenko et al. (2015) lack spatial resolution to be used for that purpose. Besides, in the vicinity of the inlet the applicability of standard procedure used to organise a reference signal for BBLIF measurements may seem controversial and requires validation itself. BBLIF may also be vulnerable to total internal reflections of fluorescence at the steep slopes of the interface; thus an alternative independent method is desirable.

Apart from BBLIF, spatiotemporal measurements of film thickness are possible with light absorption technique (Clark et al. 2001) or near-infrared attenuation method (Dupont et al. 2015), which are based on similar physical principles; thus, problems of the same kind could be expected. Planar Laser-Induced Fluorescence (PLIF) technique uses the same physical phenomenon as BBLIF, but reconstructs the film thickness in a different manner (see Schubring et al. 2010, Farias et al. 2012, Zadrazil et al. 2014, Charogiannis et al. 2015). According to Häber et al. (2015), when this method is applied to film flows in pipes, it is vulnerable to total internal reflection of fluorescence on film surface between the laser sheet and the camera, leading to overestimation of local film thickness. A method to avoid such an overestimation, proposed by Charogiannis et al. (2017) and used recently by van Eeckevelde et al. (2018), yields essential decrease in spatial resolution, which is crucial for the present experiments. Another optical method is the shadow technique. This method was used in different configurations by Ayati et al. (2017) to study the characteristics of linear waves propagating over a thick liquid layer in stratified flow, by Pan et al. (2015) to determine the average film thickness and the amplitude of disturbance waves in annular flow; by Pham et al. (2014) to study the structure of disturbance waves and the mechanisms of liquid entrainment from the film surface at the outer wall of the pipe. A non-optical solution of the problem is to use multi-electrode conductance sensors (Belt et al. 2010), though the spatial resolution in such a case is much coarser than that of the optical techniques.

According to Zhao et al. (2013), the "circumferential coherence" - i.e., the degree of two-dimensionality, - of the disturbance waves increases downstream. Given that even the developed disturbance waves often possess a three-dimensional shape (Okada et al. 1995; Cherdantsev et al. 2014) and circumferentially non-uniform height (Belt et al. 2010; Alekseenko et al. 2012), one could expect that the process of disturbance waves formation is of high three-dimensionality and studying it in one longitudinal section of the duct - as it was done by Alekseenko et al. (2015) - may lead to misinterpretation of the physical processes. Thus, a three-dimensional study of the formation process, with film thickness measurements resolved in both longitudinal and circumferential coordinates, is required. Among the aforementioned techniques, BBLIF, light absorption, near-infrared attenuation and multi-electrode conductance probes allow for three-dimensional measurements of film thickness.

This work is a continuation of the work by Alekseenko et al. (2015). We pursue three main goals here: 1) Investigation of the process of disturbance waves formation in three-dimensional approach; 2) Reliable measurements of the properties of initial waves in the area of linear growth; 3) Comparison of the measurement results to the predicted properties of waves of maximum growth. The paper is structured as follows. Section 2 presents the details on the experimental setup and measurement techniques, including previously used BBLIF-method (in

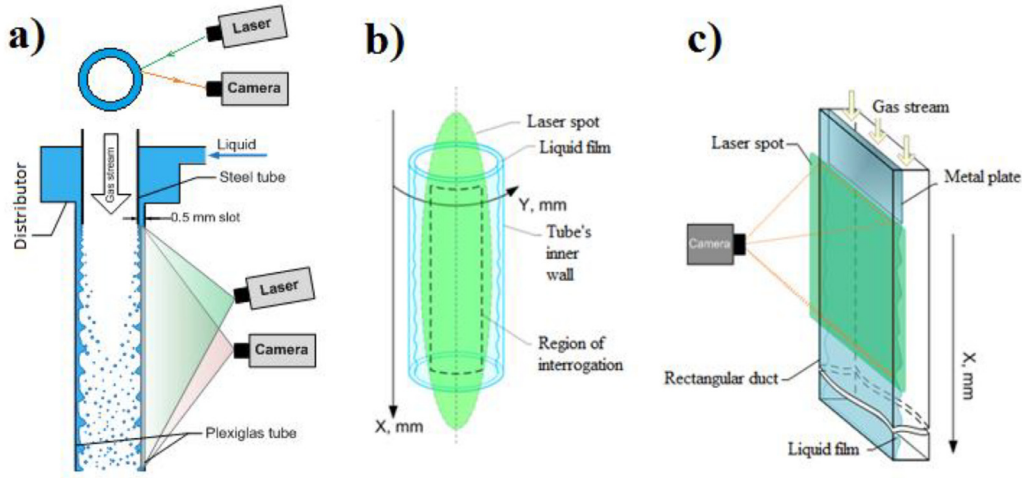


Fig. 1. Scheme of the BBLIF technique in 2D approach (a) and in 3D approach in a pipe (b) and in a rectangular duct (c).

both 2D and 3D configurations) and an independent shadow technique (2D-measurements). The applicability of the latter to the present case is considered in the Appendix using ray-tracing analysis. Section 3 gives the details of the stability theory used in the present work. Section 4 contains the measurement results and comparison to theory, whilst the last Section summarises the results.

2. Experimental details

2.1. Experimental setup and working conditions

The experiments were carried out in a circular pipe with an internal diameter of 11.7 mm and in a rectangular channel with a cross-section of $50 \times 5 \text{ mm}^2$ (Fig. 1). The liquid was fed to the walls of the channel through a tangential slot with thickness of 0.5 mm. In the rectangular duct, the film flowed along one of the wider walls, and in the circular pipe it covered the entire inner surface of the tube. In both cases the wetted perimeter was approximately the same. The advantages of a rectangular duct include the ability to observe the entire surface of the film (in the channel with a circular cross section only 1/4 of the perimeter can be studied without significant distortions), as well as the absence of optical distortions associated with the curvature of the channel. The main disadvantage of the rectangular duct is the influence of the side walls on the shape and properties of the disturbance waves.

In the present paper, adiabatic downward co-current air-water flow at near-atmospheric pressure is studied. The superficial gas velocity, V_G , varied in the range from 15 to 57 m/s with increment of 7 m/s. The Reynolds numbers of the liquid, Re_L , varied from 140 to 300; Re_L was defined as $Q/\pi D \nu$ and $Q/L \nu$ for the circular and the rectangular ducts, respectively. Here Q is the volumetric liquid flow rate, D is the internal diameter of the circular pipe, L is the width of the rectangular duct, and ν is the kinematic viscosity of the liquid (for water this value is $\sim 1.15 \cdot 10^{-6} \text{ m}^2/\text{s}$ at an operating temperature of 16 °C). According to Alekseenko et al. (2015), annular flow pattern with disturbance waves is observed in the considered conditions. The combination with the lowest gas and liquid flow rates is close to the first appearance of disturbance waves, which become well-pronounced when either liquid or gas flow rate is increased.

A detailed study of the evolution of a liquid film requires the use of experimental techniques that allow obtaining an instantaneous shape of film surface (i.e., field measurements of local film

thickness) with a high spatial and temporal resolution. Such methods include the BBLIF method and the shadow method.

2.2. The BBLIF technique

This method can measure the local film thickness being resolved either in the longitudinal (two-dimensional approach, Fig. 1a) or in both longitudinal and transverse coordinates (three-dimensional approach, Fig. 1b, c) with high spatial resolution (0.04–0.2 mm/pixel in the present study) and high sampling frequency (1–10 kHz in the present study). A continuous 2 W laser with a wavelength of 532 nm was used for excitation of fluorescence. In the 2D-case, the laser beam was transformed into a vertical laser sheet; in the 3D-configuration it was spread over an area. As a fluorescent dye, Rhodamine-6G was used at a concentration of 10 mg ($\sim 20 \mu\text{mol}$) per litre. At such concentrations, the effect of the dye on surface tension of the solution is weak (see Fig. 3 in Seno et al. 2001).

The dye absorbs laser radiation and reemits it at a larger wavelength. Local brightness of fluorescence is measured by a high-speed camera equipped with an optical filter to cut-off the reflected / scattered laser light. Based on the local brightness the local film thickness can be reconstructed using the relation:

$$I(x, y) = C(x, y) \cdot [1 - e^{-\alpha h(x, y)}] \cdot [1 + K \cdot e^{-\alpha h(x, y)}] + D(x, y)$$

Here $I(x, y)$ is the fluorescence intensity, $h(x, y)$ is the film thickness, x is the longitudinal coordinate, y is the transverse / circumferential coordinate, $C(x, y)$ is the compensation matrix, $D(x, y)$ is the dark level of the camera, α is the absorption coefficient, and K is the reflection coefficient of the laser light from the liquid / gas interface. Compensating matrix $C(x, y)$ is used to relate the arbitrary values of brightness to film thickness, as well as to eliminate distortions associated with uneven illumination of the working area by the laser spot, and with the curvature of the channel. In earlier experiments, the matrix C was determined during the calibration, using a thin ($Re_L = 30$) film falling under the action of gravity ($V_G = 0$) as a reference signal, $I_0(x, y)$. At low flow rates the average film thickness in such a flow is close to the Nusselt prediction, H_N . More detailed description of the LIF methodology is given in Alekseenko et al. (2012).

However, in the vicinity of the inlet the liquid flow is rearranged from a symmetrical velocity profile inside the 0.5-mm slot to the semi-parabolic velocity profile on the wall. Within this stage of hydrodynamic stabilisation, the local average film thickness is expected to gradually change from $H_s = 0.5 \text{ mm}$ to H_N .

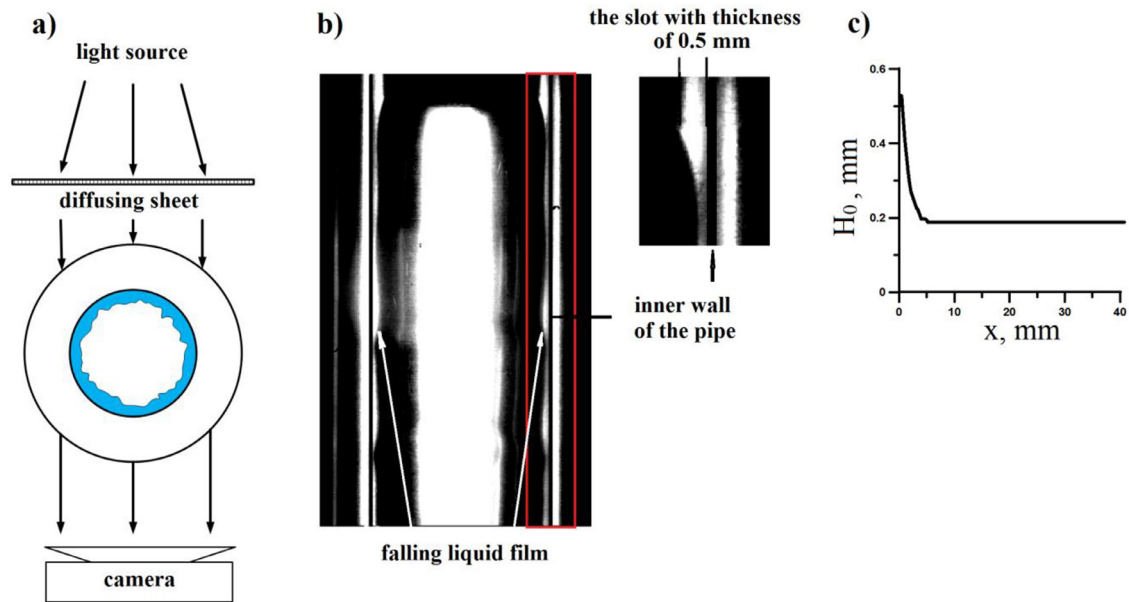


Fig. 2. a) Scheme of the shadow method; b) The image of the working section of the pipe; c) Time-averaged profile of film thickness at $Re_L=30$, $V_G=0$ m/s.

Stucheli and Ozisik (1976), Tekis et al. (1984) calculated the length of the stage of hydrodynamic stabilisation of inclined liquid films at small angles to horizon; the obtained data were in good agreement with experiments. However, the use of a similar calculation method for vertical falling films gives a difference between theoretical and experimental data (Aleksenko et al. 1994). Therefore, usage of the theoretically obtained profile of the film thickness as a reference value is not recommended. Instead, the film thickness profile should be obtained from independent experiments. The latter were conducted using the shadow technique.

2.3. The shadow method

Fig. 2a illustrates the scheme of the shadow method: white light from a 0.5 kW source passes through a diffusing screen that creates an approximately uniform field of illumination, and the camera measures the local intensity of the light passing through the channel. The main idea of the method is that the rays crossing the gas-liquid interface under low angles (which is true for the areas of interest such as shown by the red rectangle in Fig. 2b) undergo total internal reflection and do not contribute into the brightness of the corresponding part of the camera image, thus forming a shadow of the interface. On the other hand, the rays passing through the liquid, do not undergo massive reflection / refraction and create a bright image of liquid in a longitudinal section of the pipe, orthogonal to the “light source - camera” line. Examples of the bright profiles of film thickness can be seen in Fig. 2(b). In the upper part of this image the liquid in the inlet slot is also visible. Justification of this method of measurements requires a ray-tracing analysis for the light passing from the light source to the camera; the information obtained will help to accurately determine the position of the inner wall and the gas-liquid interface in the image. This analysis is presented in the Appendix.

To obtain the reference film thickness profile for the BBLIF method, time-averaged local film thickness was measured (for the case of $Re_L=30$ and $V_G=0$ m/s the result is shown in Fig. 2c) based on the radial dimension of the instantaneous film image, taking into account the distortions caused by the pipe's curvature.

Apart from that purpose, the shadow method can be used as a self-standing measurement technique: it can obtain instantaneous profiles of liquid film in longitudinal direction with high sam-

pling frequency and thus provide spatiotemporally-resolved measurements of film thickness. Unlike BBLIF, it is not vulnerable to optical distortions related to steep slopes of the interface and does not require a reference signal. On the other hand, it is limited by two-dimensional measurements. Moreover, it may underestimate the local film thickness in case of presence of three-dimensional waves; though, as it will be shown in Section 4.1, the initial waves we are going to investigate, are essentially two-dimensional. Fig. 3 shows how the raw film thickness profile obtained with the shadow technique changes with time (with step 5 ms) at different gas velocities in the region shown by the red rectangle in Fig. 2b. It can be seen from the presented images that at a large liquid flow rate at the exit from the slot (red line in Fig. 3), the liquid film is noticeably thicker than further downstream and even slightly exceeds the slot thickness. The size of this “stabilisation zone” is substantially reduced with increasing gas velocity.

The automatic interface detection algorithm, based on the Canny method of edge detection (Canny 1986), was used to determine an instantaneous film thickness profile (Figs. 4a and b, the white line corresponds to the film surface). Additional criteria were introduced to prevent detection of false boundaries: the surface of the film was considered continuous and for the case when several boundaries were detected for the same x -value, the one closest to the wall was selected.

The image size was recalculated into the dimensional film thickness values using the slot width (0.5 mm) as a linear scaling factor. The justification of the linear dependence of the actual film thickness on the transverse image size was obtained by ray-tracing analysis (see the Appendix). It was found that the transverse size of the image exceeds the actual film thickness 1.33 times, which is related to the channel curvature. Thus, spatial resolution of 0.04 mm/pixel in longitudinal direction, used in the present shadow experiments, yields spatial resolution in the radial direction equal 0.03 mm/pixel. With the absolute measurement error of 1 pixel, the uncertainty in measuring the film thickness is 6% for a thick film (~ 0.5 mm) and 15% for a thin film (~ 0.2 mm). It was also found that there is no “blind zone” inside the liquid film near neither gas-liquid nor wall-liquid boundary. Combining the instantaneous profiles of film thickness, we can construct spatiotemporal matrices of film thickness, $h(x,t)$, for each regime point. Figs. 4(c) and (d) show example fragments of $h(x,t)$ matrices (the left border

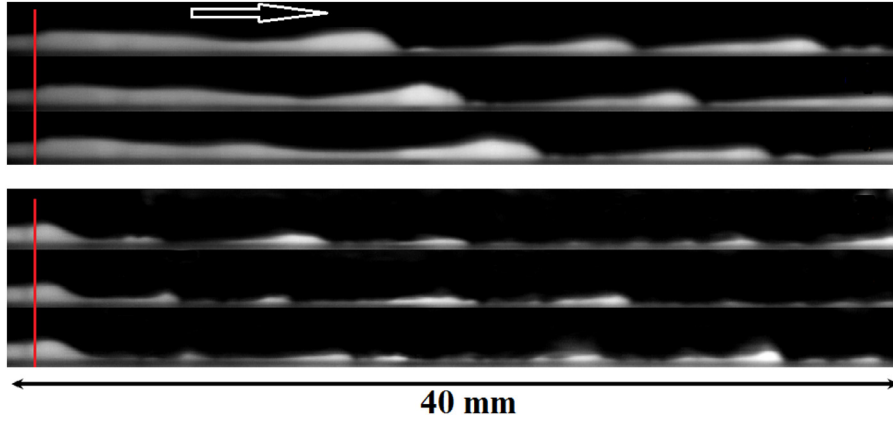


Fig. 3. Evolution of instantaneous film profiles. $Re_L=140$. Top: $V_G=15$ m/s, bottom: $V_G=36$ m/s.

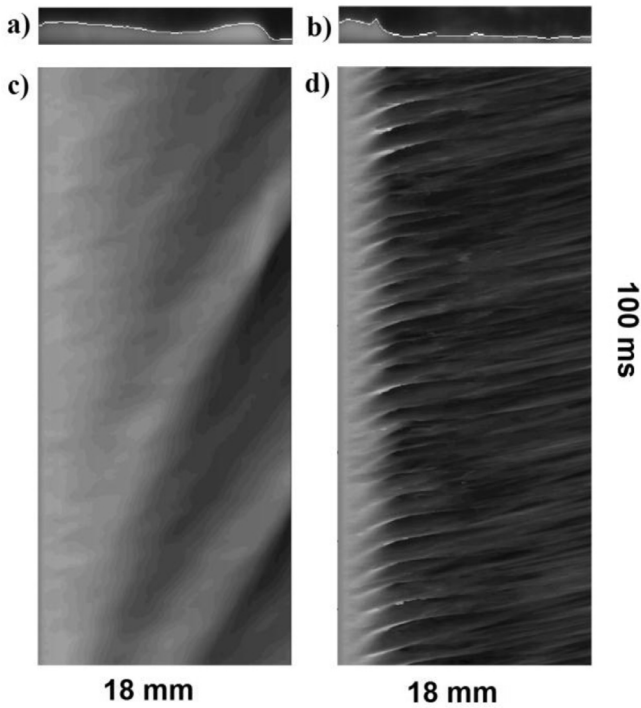


Fig. 4. (a, b) Interface detection at the instantaneous shadow images of liquid film; (c, d) Spatiotemporal records of film thickness obtained with the shadow technique. (a, c) $Re_L=140$, $V_G=15$ m/s; (b, d) $Re_L=300$, $V_G=36$ m/s.

of each image corresponds to the slot exit). The local brightness of each pixel of these images is linearly proportional to the local film thickness measured at a given point; white corresponds to the film thickness equal 1 mm.

Thus, the application of the two experimental techniques (two-dimensional approach of the BBLIF method and the shadow method) will make it possible to compare the data on the thickness of the liquid film near the entrance and to obtain reliable information on the characteristics of the initial waves. The three-dimensional approach of the BBLIF method will be used to study the waves interaction and formation of disturbance waves.

3. Theory description

3.1. Modelling the primary instability of a gas-sheared liquid film

The conjugated flow of wavy liquid film and turbulent gas stream is a well-known problem in fluid dynamics and thermo-

physics (Alekseenko and Nakoryakov, 1995). The full mathematical formulation is the system of Navier-Stokes and continuity equations for each of the phases with the conditions of no-slip and no-penetration on the channel walls, as well as kinematic and dynamic conditions at the interface. The equations may be written in dimensionless form as follows:

$$\begin{aligned} \frac{\partial \tilde{u}_f}{\partial \tilde{t}} + \tilde{u}_f \frac{\partial \tilde{u}_f}{\partial \tilde{x}} + \tilde{v}_f \frac{\partial \tilde{u}_f}{\partial \tilde{y}_f} \\ = -\frac{\partial \tilde{p}_f}{\partial \tilde{x}} + \frac{1}{\varepsilon_f Re_f} \left(\frac{\partial^2 \tilde{u}_f}{\partial \tilde{y}_f^2} + \varepsilon_f^2 \frac{\partial^2 \tilde{u}_f}{\partial \tilde{x}^2} \right) + \frac{1}{\varepsilon_f Fr} \end{aligned} \quad (1)$$

$$\begin{aligned} \varepsilon_f^2 \left[\frac{\partial \tilde{v}_f}{\partial \tilde{t}} + \tilde{u}_f \frac{\partial \tilde{v}_f}{\partial \tilde{x}} + \tilde{v}_f \frac{\partial \tilde{v}_f}{\partial \tilde{y}_f} \right] \\ = -\frac{\partial \tilde{p}_f}{\partial \tilde{y}_f} + \frac{\varepsilon_f}{Re_f} \left(\frac{\partial^2 \tilde{v}_f}{\partial \tilde{y}_f^2} + \varepsilon_f^2 \frac{\partial^2 \tilde{v}_f}{\partial \tilde{x}^2} \right) \end{aligned} \quad (2)$$

$$\frac{\partial \tilde{u}_f}{\partial \tilde{x}} + \frac{\partial \tilde{v}_f}{\partial \tilde{y}_f} = 0 \quad (3)$$

$$\begin{aligned} \frac{\partial \tilde{u}_g}{\partial \tilde{t}} + \tilde{u}_g \frac{\partial \tilde{u}_g}{\partial \tilde{x}} + \tilde{v}_g \frac{\partial \tilde{u}_g}{\partial \tilde{y}_g} = -\frac{\partial \tilde{p}_g}{\partial \tilde{x}} + \frac{1}{\varepsilon_g Re_g} \left(\frac{\partial^2 \tilde{u}_g}{\partial \tilde{y}_g^2} + \varepsilon_g^2 \frac{\partial^2 \tilde{u}_g}{\partial \tilde{x}^2} \right) \\ - \left(\frac{\partial \langle \tilde{u}'_g \tilde{u}'_g \rangle}{\partial \tilde{x}} + \frac{\partial \langle \tilde{u}'_g \tilde{v}'_g \rangle}{\partial \tilde{y}_g} \right) \end{aligned} \quad (4)$$

$$\begin{aligned} \varepsilon_g^2 \left[\frac{\partial \tilde{v}_g}{\partial \tilde{t}} + \tilde{u}_g \frac{\partial \tilde{v}_g}{\partial \tilde{x}} + \tilde{v}_g \frac{\partial \tilde{v}_g}{\partial \tilde{y}_g} \right] = -\frac{\partial \tilde{p}_g}{\partial \tilde{y}_g} + \frac{\varepsilon_g}{Re_g} \left(\frac{\partial^2 \tilde{v}_g}{\partial \tilde{y}_g^2} + \varepsilon_g^2 \frac{\partial^2 \tilde{v}_g}{\partial \tilde{x}^2} \right) \\ - \varepsilon_g^2 \left(\frac{\partial \langle \tilde{u}'_g \tilde{v}'_g \rangle}{\partial \tilde{x}} + \frac{\partial \langle \tilde{v}'_g \tilde{v}'_g \rangle}{\partial \tilde{y}_g} \right) \end{aligned} \quad (5)$$

$$\frac{\partial \tilde{u}_g}{\partial \tilde{x}} + \frac{\partial \tilde{v}_g}{\partial \tilde{y}_g} = 0 \quad (6)$$

The boundary conditions are:

$$\tilde{u}_f|_{\tilde{y}_f=0} = \tilde{v}_f|_{\tilde{y}_f=0} = 0 \quad (7)$$

$$\tilde{v}_f|_{\tilde{y}_f=\tilde{h}} = \frac{\partial \tilde{h}}{\partial \tilde{t}} + \tilde{u}_f|_{\tilde{y}_f=\tilde{h}} \frac{\partial \tilde{h}}{\partial \tilde{x}} \quad (8)$$

$$\tilde{v}_g|_{\tilde{y}_g=\tilde{R}} = \frac{\partial \tilde{u}_g}{\partial \tilde{y}_g} \Big|_{\tilde{y}_g=\tilde{R}} = 0 \quad (9)$$

$$\tilde{u}_g|_{\tilde{y}_g=\frac{\tilde{h}-1}{d}} = \varepsilon_u \tilde{u}_f|_{\tilde{y}_f=\tilde{h}} \quad (10)$$

$$\tilde{v}_g|_{\tilde{y}_g=\frac{h-1}{d}} = \varepsilon_u \tilde{v}_f|_{\tilde{y}_f=\tilde{h}} \quad (11)$$

$$\begin{aligned} \tilde{p}_f|_{\tilde{y}_f=\tilde{h}} - \frac{2\varepsilon_f}{\text{Re}_f} \frac{\partial \tilde{v}_f}{\partial \tilde{y}_f} \Big|_{\tilde{y}_f=\tilde{h}} \left(1 - \varepsilon_f^2 \left(\frac{\partial \tilde{h}}{\partial \tilde{x}} \right)^2 \right) - \frac{\partial \tilde{h}}{\partial \tilde{x}} \left(\frac{\partial \tilde{u}_f}{\partial \tilde{y}_f} + \varepsilon_f^2 \frac{\partial \tilde{v}_f}{\partial \tilde{x}} \right) \Big|_{\tilde{y}_f=\tilde{h}} \\ 1 + \varepsilon_f^2 \left(\frac{\partial \tilde{h}}{\partial \tilde{x}} \right)^2 \\ + \frac{\text{We} \varepsilon_f^2 \frac{\partial^2 \tilde{h}}{\partial \tilde{x}^2}}{\left(1 + \varepsilon_f^2 \left(\frac{\partial \tilde{h}}{\partial \tilde{x}} \right)^2 \right)^{3/2}} = \\ = \frac{\varepsilon_\rho}{\varepsilon_u^2} \tilde{p}_g \Big|_{\tilde{y}_g=\frac{h-1}{d}} - \frac{2\varepsilon_f}{\text{Re}_f} \frac{\varepsilon_\mu}{d\varepsilon_u} \\ \frac{\varepsilon_g}{\varepsilon_f} \frac{\partial \tilde{v}_g}{\partial \tilde{y}_g} \Big|_{\tilde{y}_g=\frac{h-1}{d}} \left(1 - \varepsilon_f^2 \left(\frac{\partial \tilde{h}}{\partial \tilde{x}} \right)^2 \right) - \frac{\partial \tilde{h}}{\partial \tilde{x}} \left(\frac{\partial \tilde{u}_g}{\partial \tilde{y}_g} + \varepsilon_g^2 \frac{\partial \tilde{v}_g}{\partial \tilde{x}} \right) \Big|_{\tilde{y}_g=\frac{h-1}{d}} \\ 1 + \varepsilon_f^2 \left(\frac{\partial \tilde{h}}{\partial \tilde{x}} \right)^2 \end{aligned} \quad (12)$$

$$\begin{aligned} 4\varepsilon_f^2 \frac{\partial \tilde{h}}{\partial \tilde{x}} \frac{\partial \tilde{v}_f}{\partial \tilde{y}_f} \Big|_{\tilde{y}_f=\tilde{h}} + \left(\frac{\partial \tilde{u}_f}{\partial \tilde{y}_f} + \varepsilon_f^2 \frac{\partial \tilde{v}_f}{\partial \tilde{x}} \right) \Big|_{\tilde{y}_f=\tilde{h}} \left(1 - \varepsilon_f^2 \left(\frac{\partial \tilde{h}}{\partial \tilde{x}} \right)^2 \right) = \\ = 4 \frac{\varepsilon_\mu \varepsilon_f^2}{\varepsilon_u} \frac{\partial \tilde{h}}{\partial \tilde{x}} \frac{\partial \tilde{v}_g}{\partial \tilde{y}_g} \Big|_{\tilde{y}_g=\frac{h-1}{d}} + \frac{\varepsilon_\mu}{d\varepsilon_u} \left(\frac{\partial \tilde{u}_g}{\partial \tilde{y}_g} + \varepsilon_g^2 \frac{\partial \tilde{v}_g}{\partial \tilde{x}} \right) \Big|_{\tilde{y}_g=\frac{h-1}{d}} \\ \left(1 - \varepsilon_f^2 \left(\frac{\partial \tilde{h}}{\partial \tilde{x}} \right)^2 \right) \end{aligned} \quad (13)$$

Here the dimensionless variables were introduced by the relationships:

$$\begin{aligned} x = L\tilde{x}, \quad t = (L/U)\tilde{t}, \quad y = H\tilde{y}_f = H + y_*\tilde{y}_g, \quad u_f = U\tilde{u}_f, \\ v_f = (UH/L)\tilde{v}_f, \quad u_g = u_* (\tilde{u}_g + \tilde{u}'_g), \\ v_g = (u_* y_*/L) (\tilde{v}_g + \tilde{v}'_g), \quad p_f = \rho_f U^2 \tilde{p}_f, \quad p_g = \rho_g u_*^2 \tilde{p}_g, \quad h = H\tilde{h}, \\ R = H + y_* \tilde{R}, \quad d = y_*/H, \\ \text{Fr} = U^2/gH, \quad \text{We} = \sigma/\rho_f H U^2, \quad \text{Re}_f = \rho_f UH/\mu_f, \\ \text{Re}_g = \rho_g u_* y_*/\mu_g = 1, \quad \varepsilon_f = H/L, \end{aligned}$$

$\varepsilon_g = y_*/L$, $\varepsilon_\mu = \mu_g/\mu_f$, $\varepsilon_\phi = \phi_g/\phi_f$, $\varepsilon_u = U/u_*$. Where L is the characteristic scale over the longitudinal coordinate x , t is the time, y is the transversal coordinate from the bottom of the channel, U is the superficial fluid velocity, H is the unperturbed liquid film height, R is the half of the channel height, u_* is the shear velocity of turbulent gas flow, $y_* = \rho_g \mu_g / u_*$ is the frictional length scale, $\tilde{u}'$ and $\tilde{v}'$ are the fluctuating components of longitudinal and transversal velocities, respectively, $\tilde{u}$ and $\tilde{v}$ are their mean components, p is the pressure, ρ is the density, μ is the dynamic viscosity, g is the acceleration of gravity, σ is the liquid surface tension, and lower indexes f and g correspond to fluid and gas, respectively.

One of the most efficient ways to reduce the amount of calculations during modelling is to avoid full scale computation of the turbulent gas flow. Since the liquid moves much slower than the gas $\varepsilon_u \ll 1$, it can be considered as a motionless wavy wall while solving the problem in the gas phase. Under this assumption the kinematic conditions (10–11) turn out to be just $\tilde{u}_g|_{\tilde{y}_g=\frac{h-1}{d}} = \tilde{v}_g|_{\tilde{y}_g=\frac{h-1}{d}} = 0$ and the gas part of the problem becomes independent of the liquid one.

Furthermore, owing to the film height smallness one can assume that the corresponding perturbations of the gas flow produced by the unevenness of the liquid film are linear. For linear

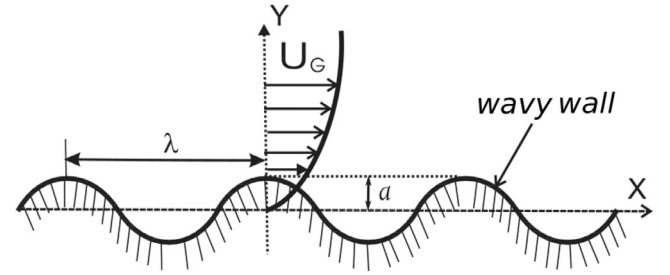


Fig. 5. Near-wall area scheme and steady-state flow velocity profile.

(sinusoidal) waves, the pressure and the shear stress perturbations at the interface may be found easily. It is worth noting that in general this approach is applicable not only to linear waves. Nonlinear perturbations of the film thickness can also be expanded into Fourier series along the longitudinal coordinate x . It allows solving the hydrodynamic equations in gas for each Fourier-harmonic independently, finding the pressure and the shear stress perturbations at the interface corresponding to each harmonic, and, summing them with corresponding multipliers to obtain the distribution of the stress tensor over the interface. After that, the governing fluid dynamics equations for the liquid should be solved numerically to calculate the film thickness and the values of all hydrodynamic variables in liquid film at the next time step. Such statement of the problem was used by, e.g., Demekhin (1981).

3.2. Stress distribution over a wavy film

Let us consider a turbulent gas flow over a rigid wavy wall. The surface of the wall has a sinusoidal shape with certain wavelength. The amplitude of the roughness a is rather small. The flow scheme is shown in Fig. 5. First, we should determine the unperturbed profile of flow velocity $\tilde{U}_g(\tilde{y}_g)$. Note that in the absence of waves the shear stress on the interface may be calculated by Blasius formula:

$$u_*^2 \equiv \frac{T}{\rho_g} = -\frac{R-H}{2\rho_g} \frac{\partial p_g}{\partial x} = \frac{\lambda}{8} u_m^2 \quad (14)$$

where u_m is the gas superficial velocity, T is the stationary shear stress, $\lambda = 0.316/\text{Re}_m^{0.25}$, and $\text{Re}_m = 2\rho_g u_m (R-H)/\mu_g$. It is known that in the vicinity of the wall the so-called logarithmic sublayer is formed in turbulent flows, beyond which the flow velocity may be considered constant. In this layer the main cause of gas motion is shear stress from the flow core, and the contribution of pressure forces is relatively weak. Neglecting it simplicity rewrite Eq. (4) for an unperturbed flow:

$$\frac{1}{\varepsilon_g} \frac{d^2 \tilde{U}_g}{d\tilde{y}_g^2} - \frac{d(\tilde{u}'_g \tilde{v}'_g)}{d\tilde{y}_g} = 0 \quad (15)$$

After a single integration obtain:

$$\frac{d\tilde{U}_g}{d\tilde{y}_g} - \varepsilon_g \langle \tilde{u}'_g \tilde{v}'_g \rangle = \tilde{T} \equiv 1 \quad (16)$$

In this work, the eddy viscosity model of turbulence is used:

$$-\varepsilon_g \langle \tilde{u}'_g \tilde{v}'_g \rangle = \frac{\mu_t}{\mu_g} \frac{d\tilde{U}_g}{d\tilde{y}_g} \quad (17)$$

$$\left(1 + \frac{\mu_t}{\mu_g} \right) \frac{d\tilde{U}_g}{d\tilde{y}_g} = 1 \quad (18)$$

with μ_t proposed by Geshev (1981) as $\mu_g/\mu_t = 2.5/\tilde{y}_g + 1015/\tilde{y}_g^3$. The solution to Eq. (18) is the following profile:

$$\begin{aligned} \tilde{U}_g(\tilde{y}_g) &= 4.33 \ln(1 + 0.091\tilde{y}_g) - 0.915 \ln(1 - 0.092\tilde{y}_g + 0.0108\tilde{y}_g^2) + \\ &+ 5.59 \arctan(0.116\tilde{y}_g - 0.492) + 2.56 \end{aligned} \quad (19)$$

Firstly, one may linearise the Navier–Stokes equations and the continuity equation taking into account that the perturbations corresponding to the wall roughness should be stationary:

$$\begin{aligned} \tilde{u}_g(\tilde{x}, \tilde{y}_g) &= \tilde{U}_g(\tilde{y}_g) + \tilde{u}_g(\tilde{x}, \tilde{y}_g), \quad \tilde{v}_g(\tilde{x}, \tilde{y}_g) = \tilde{v}_g(\tilde{x}, \tilde{y}_g), \\ \tilde{p}_g(\tilde{x}, \tilde{y}_g) &= \tilde{p}_g + \tilde{p}_g(\tilde{x}, \tilde{y}_g), \quad h(\tilde{x}) = H + h_g(\tilde{x}) = H + y_* \tilde{h}_g(\tilde{x}), \end{aligned}$$

Thus, we obtain:

$$\tilde{U}_g \frac{\partial \tilde{u}_g}{\partial \tilde{x}} + \tilde{v}_g \frac{d\tilde{U}_g}{d\tilde{y}_g} = -\frac{\partial \tilde{p}_g}{\partial \tilde{x}} + \frac{1}{\varepsilon_g} \left(\frac{\partial^2 \tilde{u}_g}{\partial \tilde{y}_g^2} + \varepsilon_g^2 \frac{\partial^2 \tilde{u}_g}{\partial \tilde{x}^2} \right) \quad (20)$$

$$\varepsilon_g^2 \tilde{U}_g \frac{\partial \tilde{v}_g}{\partial \tilde{x}} = -\frac{\partial \tilde{p}_g}{\partial \tilde{y}_g} + \varepsilon_g \left(\frac{\partial^2 \tilde{v}_g}{\partial \tilde{y}_g^2} + \varepsilon_g^2 \frac{\partial^2 \tilde{v}_g}{\partial \tilde{x}^2} \right) \quad (21)$$

$$\frac{\partial \tilde{u}_g}{\partial \tilde{x}} + \frac{\partial \tilde{v}_g}{\partial \tilde{y}_g} = 0 \quad (22)$$

Let us consider the area closely adjacent to the wavy wall. In Fig. 5 this area is presented in large scale. It is seen that the established longitudinal velocity does not vanish at the wavy wall. This is why we should carefully transfer the boundary conditions to the undisturbed level (y -coordinate equal to zero):

$$\tilde{u}_g|_{\tilde{y}_g=0} + \tilde{h}_g \frac{d\tilde{U}_g}{d\tilde{y}_g} \Big|_{\tilde{y}_g=0} = \tilde{u}_g|_{\tilde{y}_g=0} + \tilde{h}_g = 0 \quad (23)$$

$$\tilde{v}_g|_{\tilde{y}_g=0} = 0 \quad (24)$$

Numerous observations of turbulent flows indicate that perturbations in a boundary layer do not penetrate too far into the gas core. So, assumingly, at a certain sufficiently long distance from the wall all the perturbations have attenuated.

$$\tilde{v}_g|_{\tilde{y}_g \rightarrow \infty} = \tilde{u}_g|_{\tilde{y}_g \rightarrow \infty} = 0 \quad (25)$$

The Eqs. (20)–(25) constitute a well-posed boundary value problem. Its solution serves to learn about perturbations of all hydrodynamic functions caused by an unevenness of the wall. In particular, one may find perturbations of normal and shear stresses at the interface, which, as it is seen from (12)–(13), are necessary to solve the problem of liquid film flow:

$$\begin{aligned} \tilde{f}_{n,g} &= 2\varepsilon_g \left(\frac{\partial \tilde{v}_g}{\partial \tilde{y}_g} \Big|_{\tilde{y}_g=0} - \frac{\partial \tilde{h}_g}{\partial \tilde{x}} \frac{d\tilde{U}_{g,0}}{d\tilde{y}_g} \Big|_{\tilde{y}_g=0} \right) \\ -\tilde{p}_g|_{\tilde{y}_g=0} &= 2\varepsilon_g \left(\frac{\partial \tilde{v}_g}{\partial \tilde{y}_g} \Big|_{\tilde{y}_g=0} - \frac{\partial \tilde{h}_g}{\partial \tilde{x}} \right) - \tilde{p}_g|_{\tilde{y}_g=0} \end{aligned} \quad (26)$$

$$\begin{aligned} \tilde{f}_{\tau,g} &= \tilde{h}_g \frac{d^2 \tilde{U}_{g,0}}{d\tilde{y}_g^2} \Big|_{\tilde{y}_g=0} + \frac{\partial \tilde{u}_g}{\partial \tilde{y}_g} \Big|_{\tilde{y}_g=0} + \varepsilon_g^2 \frac{\partial \tilde{v}_g}{\partial \tilde{x}} \Big|_{\tilde{y}_g=0} \\ &= \frac{\partial \tilde{u}_g}{\partial \tilde{y}_g} \Big|_{\tilde{y}_g=0} + \varepsilon_g^2 \frac{\partial \tilde{v}_g}{\partial \tilde{x}} \Big|_{\tilde{y}_g=0} \end{aligned} \quad (27)$$

Let us differentiate Eq. (20) over coordinate $\tilde{y}_g$, and Eq. (21) over coordinate $\tilde{x}$, subtract the results and, thus, exclude the pressure perturbation:

$$\begin{aligned} \varepsilon_g \left(\tilde{U}_g \left(\frac{\partial^3 \tilde{v}_g}{\partial \tilde{x} \partial \tilde{y}_g^2} + \varepsilon_g^2 \frac{\partial^3 \tilde{v}_g}{\partial \tilde{x}^3} \right) - \frac{\partial \tilde{v}_g}{\partial \tilde{x}} \frac{d^2 \tilde{U}_g}{d\tilde{y}_g^2} \right) \\ = \frac{\partial^4 \tilde{v}_g}{\partial \tilde{y}_g^4} + 2\varepsilon_g^2 \frac{\partial^4 \tilde{v}_g}{\partial \tilde{x}^2 \partial \tilde{y}_g^2} + \varepsilon_g^4 \frac{\partial^4 \tilde{v}_g}{\partial \tilde{x}^4} \end{aligned} \quad (28)$$

Solutions to this equation have the form of plane waves $\tilde{v}_g(\tilde{x}, \tilde{y}_g) = \tilde{\alpha} \tilde{h}_g \hat{v}_g(\tilde{y}_g) e^{i\tilde{\alpha}\tilde{x}} + \tilde{\alpha} \tilde{h}_g^* \hat{v}_g^*(\tilde{y}_g) e^{-i\tilde{\alpha}\tilde{x}}$, $\tilde{h}_g(\tilde{x}) = \hat{h}_g e^{i\tilde{\alpha}\tilde{x}} + \hat{h}_g^* e^{-i\tilde{\alpha}\tilde{x}}$

where $\hat{v}_g$ is a complex amplitude of the Fourier harmonic, and the upper asterisk is an operator of complex conjugation. Substituting this in (28) considering boundary conditions (23)–(25) obtain the Orr–Sommerfeld equation:

$$\begin{aligned} i\varepsilon_g \tilde{\alpha} \left(\tilde{U}_g \left(\frac{d^2 \hat{v}_g}{d\tilde{y}_g^2} - \varepsilon_g^2 \tilde{\alpha}^2 \hat{v}_g \right) - \hat{v}_g \frac{d^2 \tilde{U}_g}{d\tilde{y}_g^2} \right) \\ = \frac{d^4 \hat{v}_g}{d\tilde{y}_g^4} - 2\varepsilon_g^2 \tilde{\alpha}^2 \frac{d^2 \hat{v}_g}{d\tilde{y}_g^2} + \varepsilon_g^4 \tilde{\alpha}^4 \hat{v}_g \end{aligned} \quad (29)$$

$$\hat{v}_g(0) = 0 \quad (30)$$

$$\frac{d\hat{v}_g}{d\tilde{y}_g}(0) = i \frac{d\tilde{U}_g}{d\tilde{y}_g}(0) = i \quad (31)$$

The remaining pair of conditions corresponds to the area quite remote from the wall. A trivial way to set boundary conditions at such a distance is to assume that the amplitude of the transversal velocity perturbation $\hat{v}_g$ and its derivative vanish. Another method described by Goldshtic and Shtern (1977) appears to be more accurate for such a problem. It is based on the assumption that at large distances the unperturbed velocity becomes constant. In this case the equation can be solved analytically, and the perturbation $\hat{v}_g$, decreasing at infinity, is a sum of two exponents. Indeed, let us rewrite the Eq. (29) assuming a constant profile:

$$\left(\frac{d^2}{d\tilde{y}_g^2} - \varepsilon_g^2 \tilde{\alpha}^2 \right) \left(\frac{d^2 \hat{v}_g}{d\tilde{y}_g^2} - (\varepsilon_g^2 \tilde{\alpha}^2 + i\varepsilon_g \tilde{\alpha} \tilde{U}_g(\tilde{l})) \hat{v}_g \right) = 0 \quad (32)$$

The solution is now obvious:

$$\hat{v}_g(\tilde{y}_g) = c_1 e^{-\varepsilon_g \tilde{\alpha} \tilde{y}_g} + c_2 e^{-\gamma \tilde{y}_g}, \quad \gamma = \sqrt{\varepsilon_g^2 \tilde{\alpha}^2 + i\varepsilon_g \tilde{\alpha} \tilde{U}_g(\tilde{l})} \quad (33)$$

It is convenient to introduce new variables:

$$\theta = \frac{d^2 \hat{v}_g}{d\tilde{y}_g^2} - \varepsilon_g^2 \tilde{\alpha}^2 \hat{v}_g = (\gamma^2 - \varepsilon_g^2 \tilde{\alpha}^2) c_2 e^{-\gamma \tilde{y}_g}, \quad S = \frac{d\theta}{d\tilde{y}_g} = -\gamma \theta \quad (34)$$

$$Q = \frac{d\hat{v}_g}{d\tilde{y}_g}, \quad \hat{v}_g + Q = c_2 (\varepsilon_g \tilde{\alpha} - \gamma) e^{-\gamma \tilde{y}_g} = -\frac{\theta}{\varepsilon_g \tilde{\alpha} + \gamma} \quad (35)$$

and, finally, to obtain the boundary conditions at sufficiently large distance $\tilde{l}$:

$$S(\tilde{l}) = -\gamma \theta(\tilde{l}), \quad \varepsilon_g \tilde{\alpha} \hat{v}_g(\tilde{l}) + Q(\tilde{l}) = -\frac{\theta(\tilde{l})}{\varepsilon_g \tilde{\alpha} + \gamma} \quad (36)$$

To solve Orr–Sommerfeld equation numerically we have chosen the variant of Ricatti method described by Goldshtic and Shtern (1977). The method is closely connected with a three-diagonal algorithm for the second-order differential equations but may be used for equations of any order. Orr–Sommerfeld Eq. (29) is converted into a system in a usual way, so the problem takes the form:

$$\begin{aligned} \frac{d\hat{v}_g}{d\tilde{y}_g} &= Q, \quad \frac{dQ}{d\tilde{y}_g} = \theta + \varepsilon_g^2 \tilde{\alpha}^2 \hat{v}_g, \quad \frac{d\theta}{d\tilde{y}_g} = S, \\ \frac{dS}{d\tilde{y}_g} &= (\varepsilon_g^2 \tilde{\alpha}^2 + i\varepsilon_g \tilde{\alpha} \tilde{U}_g) \theta - i\varepsilon_g \tilde{\alpha} \frac{d^2 \tilde{U}_g}{d\tilde{y}_g^2} \hat{v}_g \end{aligned} \quad (37)$$

$$\begin{aligned} \hat{v}_g(0) &= 0, \quad Q(0) = i, \quad S(\tilde{l}) = -\gamma \theta(\tilde{l}), \\ Q(\tilde{l}) &= -\varepsilon_g \tilde{\alpha} \hat{v}_g(\tilde{l}) - \frac{\theta(\tilde{l})}{\varepsilon_g \tilde{\alpha} + \gamma} \end{aligned} \quad (38)$$

Representing the solution in the form of equality constraints:

$$\begin{pmatrix} \hat{v}_g(\tilde{y}_g) \\ Q(\tilde{y}_g) \end{pmatrix} = \begin{pmatrix} B_{11}(\tilde{y}_g) & B_{12}(\tilde{y}_g) \\ B_{21}(\tilde{y}_g) & B_{22}(\tilde{y}_g) \end{pmatrix} \begin{pmatrix} \theta(\tilde{y}_g) \\ S(\tilde{y}_g) \end{pmatrix} + \begin{pmatrix} U_1(\tilde{y}_g) \\ U_2(\tilde{y}_g) \end{pmatrix} \quad (39)$$

and noting that boundary conditions for $\hat{v}_g(0)$ and $Q(0)$ should be satisfied regardless of the values of $\theta(0)$ and $S(0)$, we find initial conditions for the coefficients $B_{ij}(0)$ and $U_i(0)$:

$$B_{ij}(0) = 0, \quad U_1(0) = 0, \quad U_2(0) = i \quad (40)$$

Differentiate Eq. (39) and substitute linear expressions (39) as functions $\hat{v}_g$, Q of independent variables θ , S in the resultant equations. Equalizing the coefficients at independent variables in the so found equations obtain the system of differential equations for the functions $B_{ij}(\tilde{y}_g)$ and $U_i(\tilde{y}_g)$:

$$\frac{dB_{11}}{d\tilde{y}_g} = B_{21} + B_{12} \left(i\varepsilon_g \tilde{\alpha} \frac{d^2 \tilde{U}_g}{d\tilde{y}_g^2} B_{11} - (i\varepsilon_g \tilde{\alpha} \tilde{U}_g + \varepsilon_g^2 \tilde{\alpha}^2) \right) \quad (41)$$

$$\frac{dB_{12}}{d\tilde{y}_g} = B_{22} - B_{11} + i\varepsilon_g \tilde{\alpha} \frac{d^2 \tilde{U}_g}{d\tilde{y}_g^2} B_{12}^2 \quad (42)$$

$$\frac{dB_{21}}{d\tilde{y}_g} = 1 + \varepsilon_g^2 \tilde{\alpha}^2 B_{11} + B_{22} \left(i\varepsilon_g \tilde{\alpha} \frac{d^2 \tilde{U}_g}{d\tilde{y}_g^2} B_{11} - (i\varepsilon_g \tilde{\alpha} \tilde{U}_g + \varepsilon_g^2 \tilde{\alpha}^2) \right) \quad (43)$$

$$\frac{dB_{22}}{d\tilde{y}_g} = B_{12} \left(i\varepsilon_g \tilde{\alpha} \frac{d^2 \tilde{U}_g}{d\tilde{y}_g^2} B_{22} + \varepsilon_g^2 \tilde{\alpha}^2 \right) - B_{21} \quad (44)$$

$$\frac{dU_1}{d\tilde{y}_g} = U_2 + i\varepsilon_g \tilde{\alpha} \frac{d^2 \tilde{U}_g}{d\tilde{y}_g^2} B_{12} U_1 \quad (45)$$

$$\frac{dU_2}{d\tilde{y}_g} = i\varepsilon_g \tilde{\alpha} \frac{d^2 \tilde{U}_g}{d\tilde{y}_g^2} B_{22} U_1 + \varepsilon_g^2 \tilde{\alpha}^2 U_1 \quad (46)$$

Eq. (39) together with boundary conditions (38) at rather long distance $\tilde{l}$ constitute a fourth order system of linear equations, whose solution gives all necessary information to start the back substitution process. The latter pair of Eq. (38) is integrated with the use of linear Eq. (39) in which the matrix $B_{ij}(\tilde{y}_g)$ and the vector $U_i(\tilde{y}_g)$ are already known. At the integration of the Cauchy problem it does not matter which system to deal with (linear or nonlinear). The latter case is even more preferable, since exponential growth and oscillations are usually excluded. Finally, one may obtain the normal and the shear stresses at the interface:

$$\hat{f}_{n,g} = -\frac{S(0)}{\tilde{\alpha} \varepsilon_g} \quad (47)$$

$$\hat{f}_{\tau,g} = i\theta(0) \quad (48)$$

Here $\hat{f}_{n,g}(\tilde{x}) = \hat{h}_g \hat{f}_{n,g} e^{i\tilde{\alpha}\tilde{x}} + \hat{h}_g^* \hat{f}_{n,g}^* e^{-i\tilde{\alpha}\tilde{x}}$, $\hat{f}_{\tau,g}(\tilde{x}) = \hat{h}_g \hat{f}_{\tau,g} e^{i\tilde{\alpha}\tilde{x}} + \hat{h}_g^* \hat{f}_{\tau,g}^* e^{-i\tilde{\alpha}\tilde{x}}$.

The calculated modulus of the amplitude and phase angle of the stresses are presented in Fig. 6 and compared to similar calculations by Thorsness et al. (1978) and Tseluiko and Kalliadasis (2011). Data of Thorsness et al. were obtained using the same approach; the differences can be explained by different ways of setting the boundary conditions. Data of Tseluiko and Kalliadasis were obtained using a different approach. The main difference between the results are observed in the range of low wavenumbers. In this paper, the waves of maximum growth will be considered; their wavenumbers belong to the right-hand part of each plot. In this range good quantitative agreement between the different models is expected.

3.3. Liquid film

To simulate the dynamics of linear perturbations on the surface of a thin layer of viscous liquid sheared by a turbulent gas stream, a similar Orr–Sommerfeld equation may be derived. Steady profile of liquid velocity follows directly from Eq. (4):

$$\tilde{U}_f(\tilde{y}_f) = -\frac{\text{Re}_f}{2\text{Fr}} (\tilde{y}_f^2 - 2\tilde{y}_f) + \tilde{T}_f \tilde{y}_f \quad (49)$$

Here $T = \tilde{T}_f \frac{\mu_f U}{H}$, $\tilde{T}_f = \frac{\varepsilon_\mu}{\varepsilon_u \tilde{q}}$. Now we can determine the value of the superficial liquid velocity U :

$$\tilde{q} = \int_0^1 \tilde{U}_f d\tilde{y}_f = \frac{\text{Re}_f}{3\text{Fr}} + \frac{\tilde{T}_f}{2} = 1 \quad (50)$$

$$U = \frac{\rho_f g H^2}{3\mu_f} + \frac{\varepsilon_\mu u_*}{2d} \quad (51)$$

To carry out a linear stability analysis with respect to small perturbations, a linearisation procedure is performed:

$$\begin{aligned} \tilde{u}_f(\tilde{x}, \tilde{y}_f, \tilde{t}) &= \tilde{U}_f(\tilde{y}_f) + \tilde{u}_f(\tilde{x}, \tilde{y}, \tilde{t}), \quad \tilde{v}_f(\tilde{x}, \tilde{y}_f, \tilde{t}) = \tilde{v}_f(\tilde{x}, \tilde{y}_f, \tilde{t}) \\ \tilde{p}_f(\tilde{x}, \tilde{y}_f, \tilde{t}) &= \tilde{p}_f(\tilde{x}, \tilde{y}_f, \tilde{t}), \quad \tilde{h}(\tilde{x}, \tilde{t}) = 1 + \tilde{h}(\tilde{x}, \tilde{t}) \end{aligned} \quad (52)$$

As a result, we have the linear equations:

$$\frac{\partial \tilde{u}_f}{\partial \tilde{t}} + \tilde{U}_f \frac{\partial \tilde{u}_f}{\partial \tilde{x}} + \tilde{v}_f \frac{d\tilde{U}_f}{d\tilde{y}_f} = -\frac{\partial \tilde{p}_f}{\partial \tilde{x}} + \frac{1}{\varepsilon_f \text{Re}_f} \left(\frac{\partial^2 \tilde{u}_f}{\partial \tilde{y}_f^2} + \varepsilon_f^2 \frac{\partial^2 \tilde{u}_f}{\partial \tilde{x}^2} \right) \quad (53)$$

$$\varepsilon_f^2 \left[\frac{\partial \tilde{v}_f}{\partial \tilde{t}} + \tilde{U}_f \frac{\partial \tilde{v}_f}{\partial \tilde{x}} \right] = -\frac{\partial \tilde{p}_f}{\partial \tilde{y}_f} + \frac{\varepsilon_f}{\text{Re}_f} \left(\frac{\partial^2 \tilde{v}_f}{\partial \tilde{y}_f^2} + \varepsilon_f^2 \frac{\partial^2 \tilde{v}_f}{\partial \tilde{x}^2} \right) \quad (54)$$

$$\frac{\partial \tilde{u}_f}{\partial \tilde{x}} + \frac{\partial \tilde{v}_f}{\partial \tilde{y}_f} = 0 \quad (55)$$

with new boundary conditions:

$$\tilde{u}_f|_{\tilde{y}_f=0} = \tilde{v}_f|_{\tilde{y}_f=0} = 0 \quad (56)$$

$$\tilde{v}_f|_{\tilde{y}_f=1} = \frac{\partial \tilde{h}}{\partial \tilde{t}} + \tilde{U}_f(1) \frac{\partial \tilde{h}}{\partial \tilde{x}} \quad (57)$$

$$\frac{d^2 \tilde{U}_f}{d\tilde{y}_f^2} \tilde{h} + \frac{\partial \tilde{u}_f}{\partial \tilde{y}_f} \Big|_{\tilde{y}_f=1} + \varepsilon_f^2 \frac{\partial \tilde{v}_f}{\partial \tilde{x}} \Big|_{\tilde{y}_f=1} = \frac{\varepsilon_\mu}{d\varepsilon_u} \tilde{f}_{\tau,g} \quad (58)$$

$$\tilde{p}_f|_{\tilde{y}_f=1} - \frac{2\varepsilon_f}{\text{Re}_f} \left(\frac{\partial \tilde{v}_f}{\partial \tilde{y}_f} \Big|_{\tilde{y}_f=1} - \frac{\partial \tilde{h}}{\partial \tilde{x}} \frac{\partial \tilde{U}_f}{\partial \tilde{y}_f} \Big|_{\tilde{y}_f=1} \right) + \text{We} \varepsilon_f^2 \frac{\partial^2 \tilde{h}}{\partial \tilde{x}^2} = -\frac{\varepsilon_\rho}{\varepsilon_u^2} \tilde{f}_{n,g} \quad (59)$$

Let us seek a solution to Eqs. (53)–(59) in the form:

$$\begin{aligned} [\tilde{u}_f(\tilde{y}), \tilde{v}_f(\tilde{y}_f), \tilde{p}_f(\tilde{y}_f), \tilde{h}](\tilde{x}, \tilde{t}) &= \\ [\hat{u}_f(\tilde{y}), \hat{v}_f(\tilde{y}_f), \hat{p}_f(\tilde{y}_f), \hat{h}] e^{i\tilde{\alpha}(\tilde{x} - \tilde{c}\tilde{t})} + c.c. \end{aligned}$$

Excluding the pressure amplitude from (53)–(54) and using the expression for the amplitude of longitudinal velocity resulting from (55) one may easily derive the Orr–Sommerfeld problem:

$$\begin{aligned} \frac{d^4 \hat{v}_f}{d\tilde{y}_f^4} - 2\varepsilon_f^2 \tilde{\alpha}^2 \frac{d^2 \hat{v}_f}{d\tilde{y}_f^2} + \varepsilon_f^4 \tilde{\alpha}^4 \hat{v}_f \\ = -i\varepsilon_f \tilde{\alpha} \text{Re}_f \left([\tilde{c} - \tilde{U}_f(\tilde{y}_f)] \left(\frac{d^2 \hat{v}_f}{d\tilde{y}_f^2} - \varepsilon_f^2 \tilde{\alpha}^2 \hat{v}_f \right) + \hat{v}_f \frac{d^2 \tilde{U}_f}{d\tilde{y}_f^2} \right) \end{aligned} \quad (60)$$

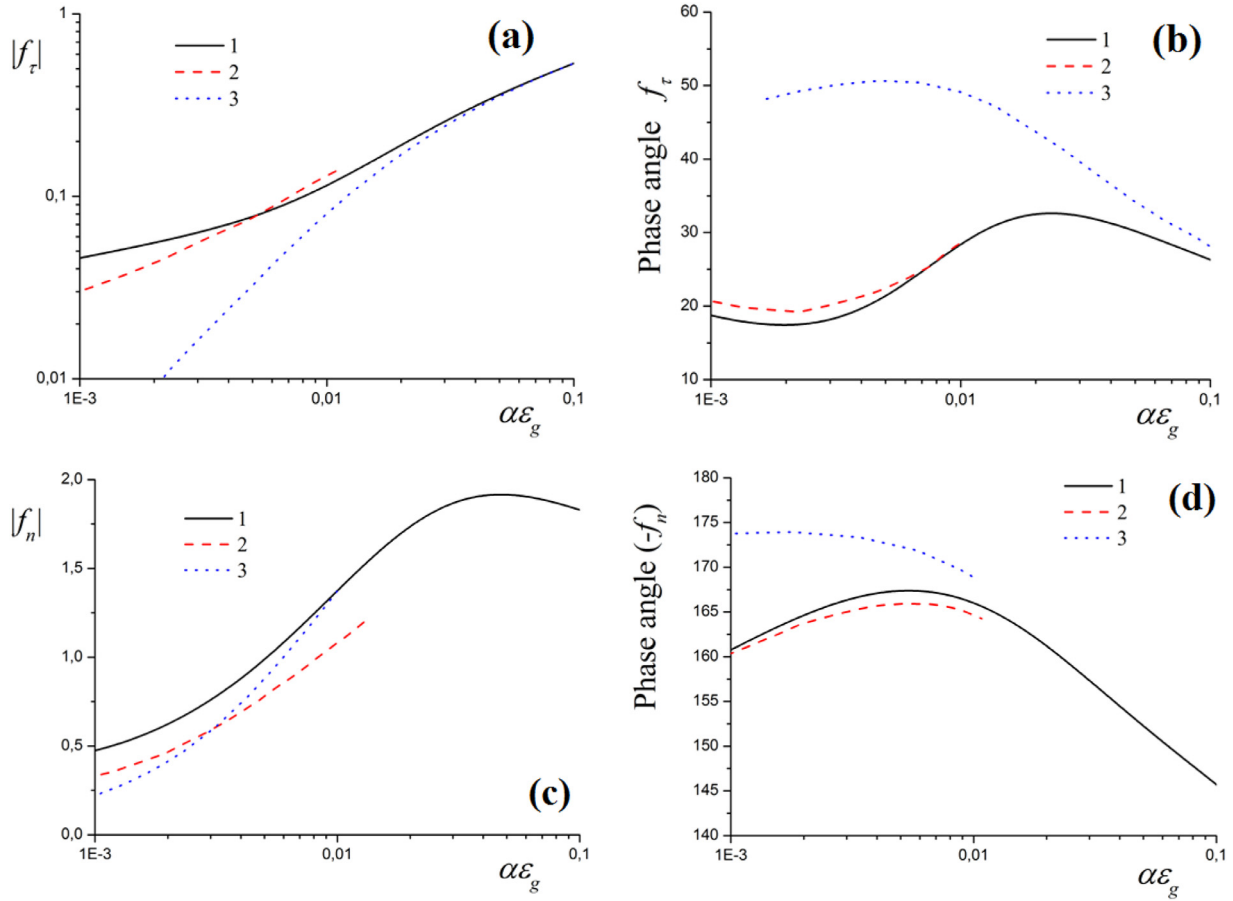


Fig. 6. The modulus of amplitude of shear stress (a) and normal stress (c), and the phase angle of shear stress (b) and normal stress (d). 1 - present model; 2 - Thorsness et al. (1978); 3 - Tseluiko and Kalliadasis (2011).

$$\hat{v}_f(0) = \frac{d\hat{v}_f}{d\hat{y}_f}(0) = 0, \quad (61)$$

$$\frac{d^2\hat{v}_f}{d\hat{y}_a^2}(1) + \tilde{a}^2\epsilon_f^2\hat{v}_f(1) + \left(\frac{d^2\tilde{U}_f}{d\hat{y}_f^2} - \frac{\epsilon_\mu}{d^2\epsilon_u}\hat{f}_{\tau,g}\right)\frac{\hat{v}_f(1)}{\tilde{c} - \tilde{U}_f(1)} = 0 \quad (62)$$

$$\begin{aligned} & \frac{d^3\hat{v}_f}{d\hat{y}_f^3}(1) + (i\epsilon_f\tilde{\alpha}\text{Re}_f(\tilde{c} - \tilde{U}_f(1)) - 3\epsilon_f^2\tilde{\alpha}^2)\frac{d\hat{v}_f}{d\hat{y}_f}(1) + \\ & + \hat{v}_f(1)\left(\left(i\epsilon_f\tilde{\alpha}\text{Re}_f - \frac{2\epsilon_f^2\tilde{\alpha}^2}{\tilde{c} - \tilde{U}_f(1)}\right)\frac{d\tilde{U}_f}{d\hat{y}_f}(1) - \frac{18}{5}\frac{i\epsilon_f\tilde{\alpha}^3\text{Re}_f}{\tilde{c} - \tilde{U}_f(1)}\right. \\ & \left. + \frac{i\epsilon_\rho\epsilon_f\tilde{\alpha}\text{Re}_f\hat{f}_{n,g}}{\epsilon_u^2d(\tilde{c} - \tilde{U}_f(1))}\right) = 0 \end{aligned} \quad (63)$$

Here, the kinematical condition at the interface $\hat{v}_f(1) = -i\tilde{\alpha}(\tilde{c} - \tilde{U}_f(1))\hat{h}$ is used. The only undetermined parameter in this problem statement is the characteristic scale of the longitudinal coordinate, which is connected with the parameter ϵ_f . It is known that at low Reynolds numbers and in the absence of gas flow the wave number of neutral perturbations (see for example Alekseenko et al. (1994)):

$$\tilde{k}_n^2 = \epsilon_f^2\tilde{\alpha}_n^2 = \frac{18}{5We} \quad (64)$$

Hence, assuming $\tilde{\alpha}_n = 1$, obtain:

$$\epsilon_f = \sqrt{\frac{18}{5We}} \quad (65)$$

In our work the expression (65) is used to consider the problem (60)–(63). In accordance with calculations α_n slightly differs from unity up to moderate values of Re_f .

The solution of eigenvalue problem (60)–(63) was carried out by the Newton method with integration over the transverse coordinate by the fourth-order Runge–Kutta method. Data on the linear stability of the undisturbed flow were obtained for fixed properties of the liquid (in this work the working fluid was water under standard conditions). In the course of the calculation, the wave number $\tilde{k}$ was considered to be real and the dependences $\tilde{c}_r(\tilde{k})$ and $\tilde{c}_{im}(\tilde{k})$ were obtained. For analysis and comparison with the experimental data, the dispersion relation $\tilde{\omega}(\tilde{k}) = \tilde{c}(\tilde{k})\tilde{k} = \tilde{\omega}_r(\tilde{k}_r) + i\tilde{\omega}_{im}(\tilde{k}_r)$ has to be inverted, having expressed $\tilde{k}(\tilde{\omega}) = \tilde{k}_r(\tilde{\omega}_r) + i\tilde{k}_{im}(\tilde{\omega}_r)$. Since $\tilde{\omega}(\tilde{k})$ was calculated only for real values of $\tilde{k}$ and because of the smallness of the ratio $\tilde{\omega}_{im}/\tilde{\omega}_r$ this work used the Taylor expansion:

$$\begin{aligned} \tilde{\omega}(\tilde{k}_r + i\tilde{k}_{im}) & \approx \tilde{\omega}(\tilde{k}_r) + i\tilde{k}_{im}\frac{d\tilde{\omega}}{d\tilde{k}}(\tilde{k}_r) \\ & \approx \tilde{\omega}_r(\tilde{k}_r) + i\tilde{\omega}_{im}(\tilde{k}_r) + i\tilde{k}_{im}\frac{d\tilde{\omega}_r}{d\tilde{k}_r}(\tilde{k}_r). \end{aligned} \quad (66)$$

Under the experimental condition $\tilde{\omega}_{im} = 0$, hence:

$$\tilde{k}_{im} = -\frac{\tilde{\omega}_{im}(\tilde{k}_r)}{\frac{d\tilde{\omega}_r}{d\tilde{k}_r}(\tilde{k}_r)} = -\frac{\tilde{c}_{im}(\tilde{k}_r)\tilde{k}_r}{\tilde{c}_r(\tilde{k}_r) + \tilde{k}_r\frac{d\tilde{c}_r}{d\tilde{k}_r}(\tilde{k}_r)}. \quad (67)$$

In addition, the dependence of the wavelength of the maximum growth of small perturbations (corresponding to the maximum of the curve $\tilde{\omega}_{im}(\tilde{k})$) on the Reynolds number of the liquid and the velocity of the gas flow was obtained.

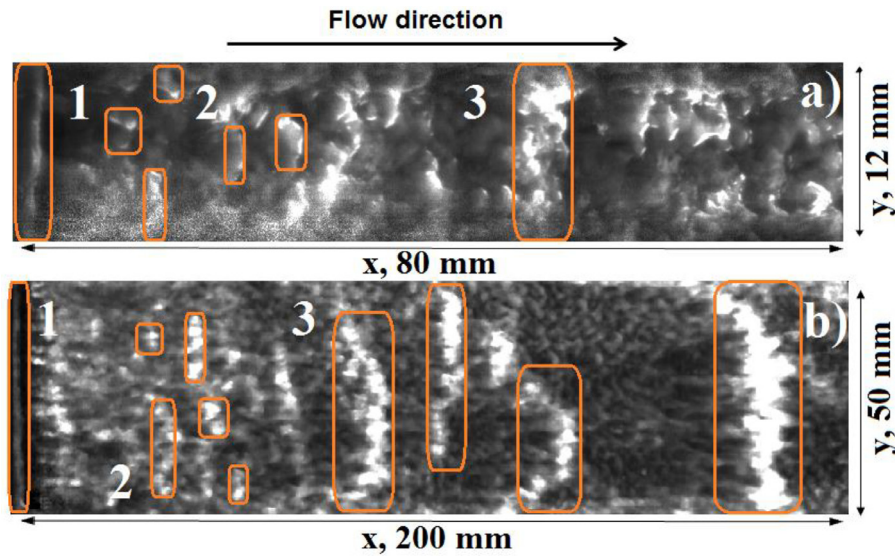


Fig. 7. Three-dimensional pattern of the film surface. $Re_L=220$, $V_G=43$ m/s. (a) cylindrical channel, (b) rectangular duct.

4. Results

4.1. Formation of disturbance waves in the 3D-approach

Using the three-dimensional LIF method, it was found that the formation of the disturbance waves occurs in several stages. The initial waves appearing near the inlet are nearly two-dimensional with straight fronts normal to the flow direction and occupying the entire width of the duct (stage 1). After some short distance these waves break into three-dimensional waves with a transverse dimension of the order of several millimetres, which propagate at different speeds (stage 2). Coalescence of these waves leads to formation of large quasi-two-dimensional disturbance waves (stage 3). Within the frames of two-dimensional approach, used by Alekseenko et al. (2015), it would have been nearly impossible to distinguish the stages 1 and 2.

Fig. 7 shows examples of instantaneous three-dimensional shape of film surface in channels with a circular ($D=11.7$ mm) and rectangular (50×5 mm²) cross-section, where the waves belonging to certain stages of evolution are marked by the respective numbers. Better details can be seen in the supplementary videos 1–4, showing time-resolved waves evolution in 3D-space.

The three-dimensional matrices $h(x,y,t)$ obtained in experiments make it possible to investigate how the film thickness varies with time at any point of the interrogation area. Comparing the temporal records obtained in different cross-sections via cross-correlation analysis, one can quantitatively describe the downstream evolution of the waves two-dimensionality. The film is considered completely two-dimensional if all the temporal records of the film thickness obtained at the same coordinate x , but different y -coordinates, y_1 and y_2 , are identical (in such a case, the normalised correlation, $b=1$). Conversely, zero correlation between such records would correspond to the complete three-dimensionality of the film surface. The method of calculating the normalised correlation is described in detail by Isaenkov et al. (2017).

The correlation field, $b(x, y_1)$, is quite uniform across y_1 -coordinate at fixed distance y_2-y_1 (equal 5 mm, see Isaenkov et al. 2017). Fig. 8 shows how the value of b , averaged across the values of y_1 , changes with x during the first 200 mm of the rectangular duct; the error bars show the standard deviation of b -data in the transverse direction. At moderate gas velocities (Fig. 8a), an appreciable peak near the inlet corresponding to the initial

two-dimensional waves (stage 1) is observed. The subsequent decrease to the values of $b \sim 0.1$ corresponds to the decay into three-dimensional waves (stage 2). During the stage 3, the transverse coherence is gradually recovered with distance, which indicates the formation of disturbance waves and increase of their transverse size. At high gas velocities, the picture is similar, but the initial waves almost immediately break down into three-dimensional waves, and the formation of disturbance waves occurs much closer to the inlet (Fig. 8b). It should be noted that the first two stages are quite similar to well-known transition to three-dimensional waves on the surface of liquid films flowing in absence of a gas flow at moderate liquid Reynolds numbers (see, e.g., Kharlamov et al. 2015). The difference between the falling and the gas-sheared films consists in the absence of the third stage, i.e., the increase of two-dimensionality at larger distances for the falling films. The small 3D-waves on falling films do not undergo irreversible coalescence in the case of contact of their side edges; thus, they do not produce the waves of larger transverse size. Similar conclusions were devised by Clark et al. (2001), who noticed that the presence of gas stream enhances probability of appearance of quasi-two-dimensional waves on film surface.

Based on dependencies $b(x)$, the boundaries between steps 1–2 and 2–3 for each flow regime can be determined. The boundaries were defined as the points where linear extrapolations of specific regions of dependencies $b(x)$ intersect the level $b=0.1$. The first region was a steep negative slope immediately after the maximum of the two-dimensionality of the initial waves. The second area was the steepest positive slope immediately after the values of $b \sim 0.1$.

The spatial length of the region of initial two-dimensional waves decreases with increasing gas velocity (Fig. 9a). The effect of liquid flow rate is the opposite: the lower the liquid flow rate, the faster the two-dimensional waves decay into three-dimensional ones, but this effect is weak in comparison with that of gas velocity. This is another difference to the case of falling films: in absence of gas stream, larger liquid Reynolds numbers yield faster loss of two-dimensionality of the waves. A possible explanation of this difference will be discussed in the end of Section 4.2.

The length of “2D-region” is only weakly affected by the shape of the duct (the whole data set can be roughly approximated by a single dependence $x \sim V_G^{-2}$, shown by the solid line in Fig. 9a). The coordinate of the beginning of the stage 3 also decreases with

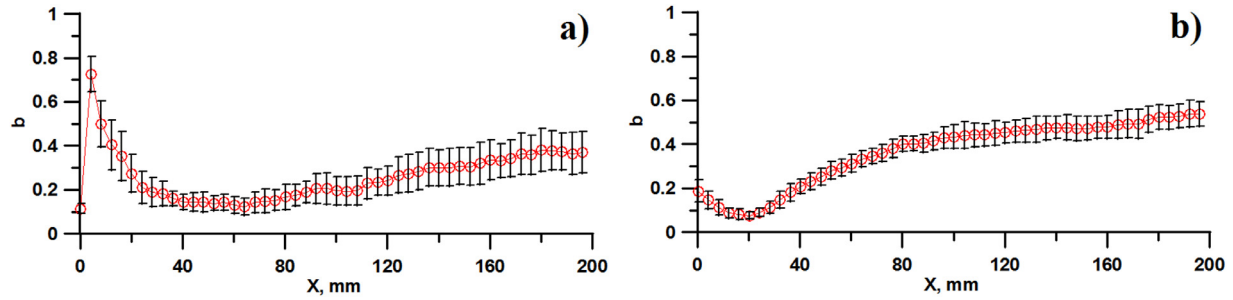


Fig. 8. The average value of the correlation along the transverse coordinate. a) $Re_L=220$, $V_G=26$ m/s; b) $Re_L=300$, $V_G=43$ m/s. Rectangular duct.

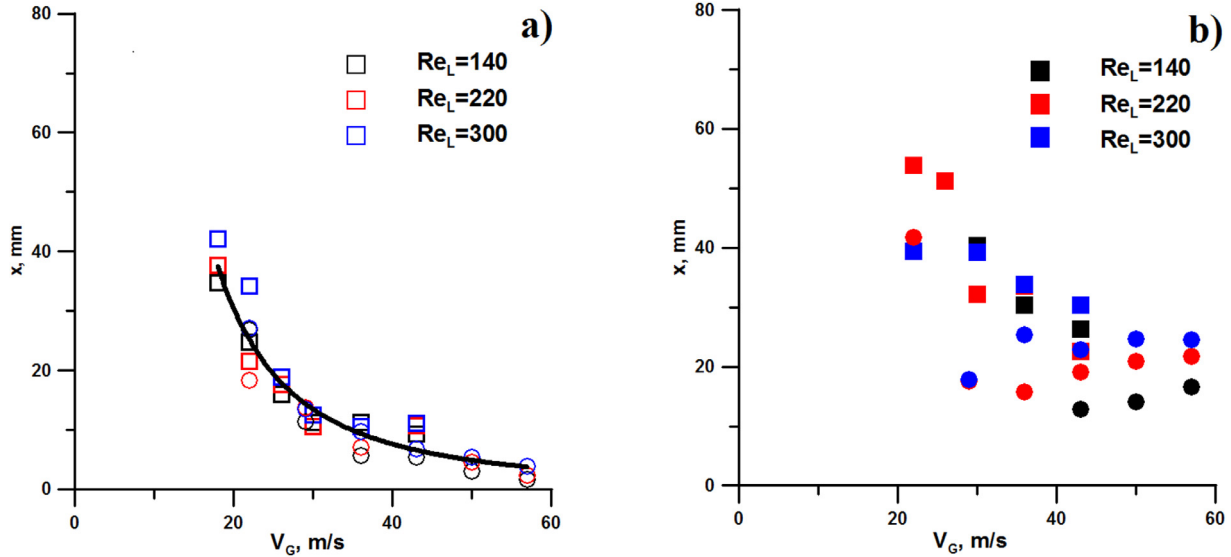


Fig. 9. a) The length of the region of initial two-dimensional waves; the solid line is the approximation of all the data points in form of $x \sim V_G^{-2}$. b) The beginning of stage 3 (formation of disturbance waves). The circles correspond to the circular pipe and the squares correspond to the rectangular duct experiments.

increasing gas velocity and increases with increasing liquid flow rate (Fig. 9b). The formation of disturbance waves begins slightly earlier in the pipe than in the rectangular duct, which can be explained by the effect of the side walls of the rectangular duct slowing the waves down. We should notice that both borders correspond to quite extreme cases, representing an upper estimate of the length of existence of initial 2D-waves and a lower estimate of the distance at which the disturbance waves become two-dimensional.

Comparison of the results of the analysis of two-dimensionality by Isaenkov et al. (2017) with the spectral structure of the waves by Alekseenko et al. (2015) showed that the initial two-dimensional waves are also of high regularity in time. The three-dimensional waves, formed as a result of destruction of the 2D-waves, are essentially irregular and are characterised by lower frequencies.

As it was mentioned in Section 2.3, the shadow method would detect only the minimum film thickness value across a range of circumferential coordinates, which would result in an underestimation of the amplitude of three-dimensional waves. The results of the 3D-experiments allow us to use the shadow method to study the properties of the initial waves in a pipe, since these waves were found to be essentially two-dimensional. On similar reasons the applicability of the shadow method in the rectangular duct is doubtful, but, since the wave patterns are quite similar in both circular and rectangular ducts, it would be enough to conduct the shadow measurements in the pipe only.

4.2. Comparison of the different measurement techniques.

Stabilisation zone length

To focus on the initial waves, the length of the working area was reduced to 40 mm in order to improve the spatial resolution in longitudinal direction to 0.04 mm/pixel. As substantiated in the Appendix, the spatial resolution in radial direction in such a case becomes 0.03 mm/pixel due to magnification of the image by the pipe curvature. The sampling frequency was set to 10 kHz to enhance the temporal resolution for frequency and velocity measurements.

Fig. 10 shows time-averaged film thickness profiles obtained with the help of the shadow method (blue line) and the BBLIF method (black line), as well as the span of standard deviation for the BBLIF-data (red lines). It can be seen from the graphs that at a low gas velocity, the thickness of the film near the inlet is noticeably higher for the BBLIF-data. The reason of this difference can be the reflection of the laser beam from the edge of the end of the metal tube, separating gas and liquid inlets, which causes additional excitation of the dye, thereby increasing the brightness of the image near the entrance. Due to the uncertainty associated with measuring the film thickness by the BBLIF method near the inlet, the growth rate of the amplitude will be determined based on the shadow measurements.

Near the entrance, the average thickness of the film has a value on the order of the gap width (0.5 mm); it decreases with increasing distance and is eventually stabilised at a constant level.

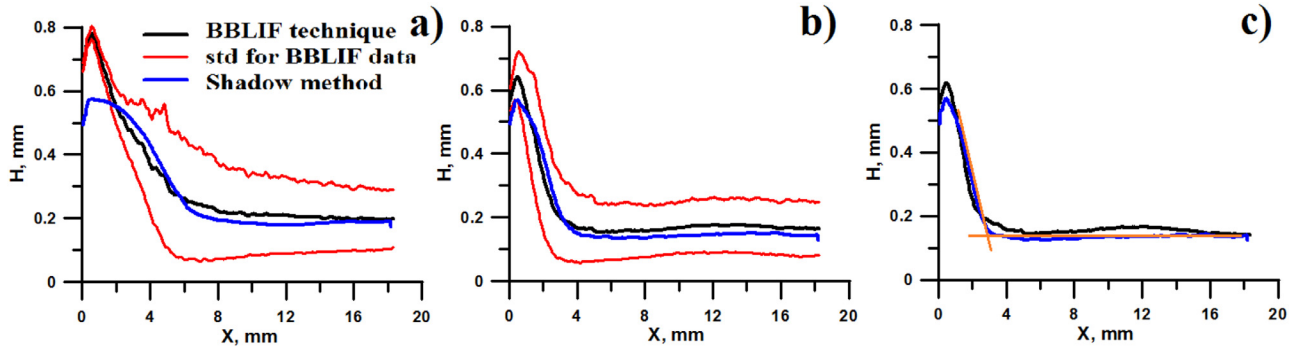


Fig. 10. Downstream change of the time-averaged film thickness, measured by BBLIF (black line) and shadow (blue line) methods. The red lines show the range of standard deviation in BBLIF measurements. The orange lines in (c) show linear approximations of the decaying slope of film thickness and stabilised average film thickness far from the inlet. $Re_L = 140$. a) $V_G = 22$ m/s; b) $V_G = 36$ m/s; c) $V_G = 43$ m/s.

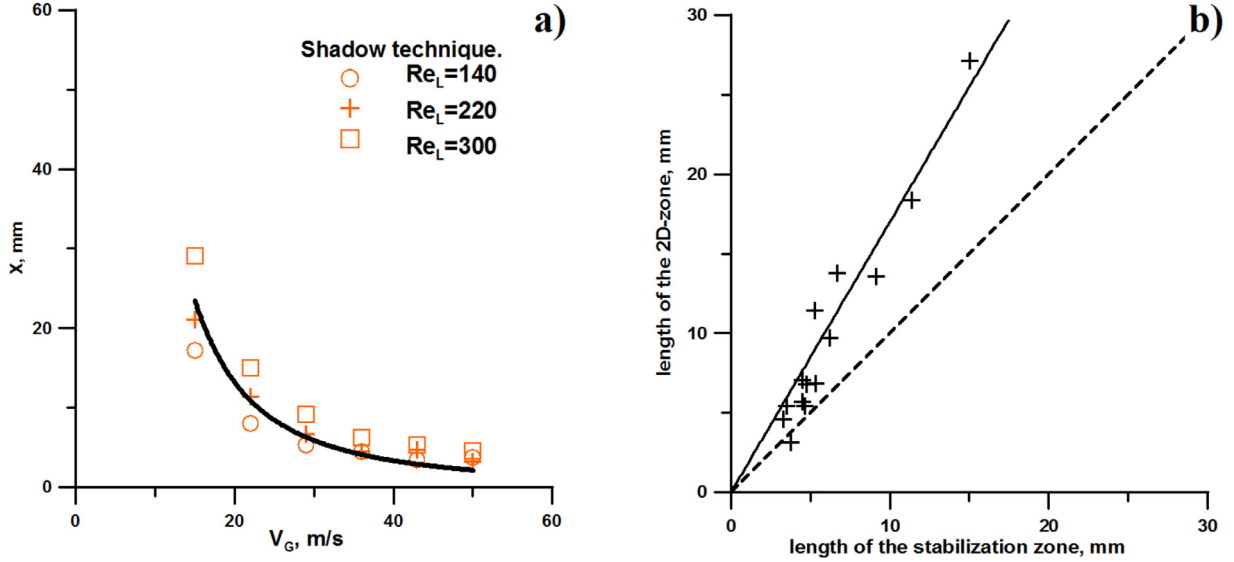


Fig. 11. a) The length of the film thickness stabilisation zone; the solid line is the approximation of all the data points in form of $x \sim V_G^{-2}$. b) Comparison of the lengths of the zones of the initial two-dimensional waves and the thick layer (the stabilisation zone). The relation $y = x$ is shown by the dashed line, and $y = 1.7x$ - by the solid line. Circular pipe experiments.

Fig. 11(a) shows the dependence of the length of the thick film ("stabilisation zone") on the flow parameters. The end of the zone was defined as the point of intersection of the stabilised value of film thickness to the linear extrapolation of the average film thickness values in the region of the maximum negative slope (see the orange lines in Fig. 10c). The length of the stabilisation zone decreases with increasing gas velocity, and increases slightly with liquid flow rate (Fig. 11a), the solid line corresponds approximation of the data in form $x \sim V_G^{-2}$. The similarity of the Figs. 9(a) and 11(a) suggests that the existence of initial two-dimensional waves is related to the thicker film stage prior to stabilisation of film thickness. Fig. 11b presents a direct comparison of the two lengths; the dashed line corresponds to the complete agreement between the two lengths, and the solid line - to the ratio of 1.7 times. Thus, the initial waves maintain some residual two-dimensionality even beyond the end of the thick layer, and their destruction into three-dimensional waves occurs after the "stabilised" value film thickness is reached. We assume that the process of destruction of two-dimensional waves is initiated at the end of the thick initial film. When the 2D-waves slide down from the thick initial film onto the thin film of "stabilised" thickness, being "assisted" by the gas flow, they receive additional perturbations and their transverse instabil-

ity may be enhanced. This hypothesis is supported by the video recordings of the film thickness, see supplementary videos 5 and 6, obtained with the shadow technique for the same conditions as the videos 3 and 4.

The interrelation between the two lengths helps to explain the qualitatively different effect of liquid Reynolds number on the length of the region of two-dimensional waves for falling and gas-sheared films. In case of falling films increase in Re_L enhances the transverse instability of the 2D-waves. In presence of a high-velocity gas stream the main source of perturbations is the turbulent gas phase, whilst the effect of Re_L is expected to be weaker. At the same time, the larger liquid flow rate provides more liquid to sustain a longer initial thick layer, where the initial waves are still two-dimensional.

4.3. Spectral structure of the initial waves

Each spatiotemporal record of film thickness $h(x,t)$ can be considered as a set of temporal records of the film thickness obtained at various downstream distances. These records can be processed using statistical and spectral analysis to extract the main properties of the waves and to investigate how these properties change

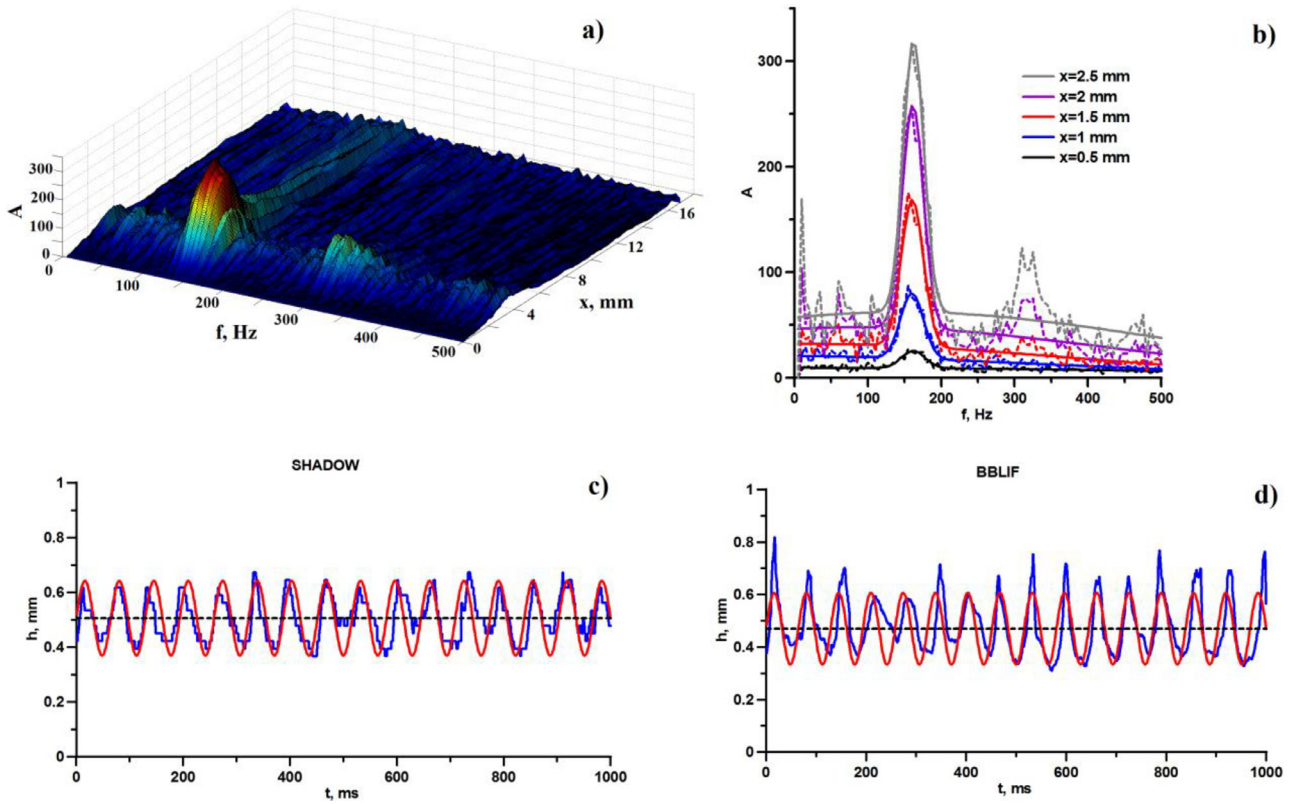


Fig. 12. (a) Downstream evolution of signal spectrum, shadow technique, $Re_1 = 140$, $V_G = 29$ m/s. (b) Spectra for 5 downstream positions along with their approximations. (c-d) Comparison of the original signal at $x = 1.5$ mm (blue line) with its sinusoidal approximation (red line) for shadow technique (c) and BBLIF (d).

with x . Each record, consisting of 10,000 points, was split into 5 parts. Fast Fourier Transform was applied to each part, yielding spectral amplitudes with frequency step of 5 Hz. The spectral amplitudes for all parts were summed up to produce the spectrum of the record.

Fig. 12(a) shows an example of downstream evolution of signal spectrum for shadow records of film thickness (first 500 Hz are shown). The first spectral amplitude, A_0 , corresponding to the mean film thickness, is omitted in this Figure. A well-pronounced hump, corresponding to the initial waves, is observed around the frequency of 160 Hz. The hump's amplitude grows rapidly for the first 3 mm and then decreases. The 1st overtone of the main hump is also distinguishable around 320 Hz, indicating the growth of nonlinearity of the signal. Signal spectra for five downstream distances from 0.5 to 2.5 mm are shown in Fig. 12(b). At this stage all the spectra may be approximated as a sum of two Gaussian functions:

$$A(f) = a_1 * \exp\left(-\frac{1}{2}\left(\frac{f - f_1}{\sigma_1}\right)^2\right) + a_2 * \exp\left(-\frac{1}{2}\left(\frac{f - f_2}{\sigma_2}\right)^2\right). \quad (68)$$

Such approximations are superimposed onto the spectra in Fig. 12(b). The first term of the approximation (68) describes the main hump of a spectrum: its expected value, f_1 , and variance, σ_1 , remain approximately the same during the regarded stage of evolution; the amplitude, a_1 , grows downstream. The second term describes the residual part of the spectrum, which is mainly represented by the noise, uniformly distributed over the frequency domain. The first “overtone” (at ~ 320 Hz) is of smaller magnitude and starts to be distinguishable later than the main hump.

The initial waves are not perfectly sinusoidal; though, their energy is spread within relatively narrow frequency range around the main frequency. The ratio σ_1/f_1 is about 10% for the whole range of investigated flow conditions. Thus, the initial waves show surprisingly high degree of regularity for a thick liquid film under strong turbulent gas shear in absence of imposed external perturbations.

Such shape of the spectrum may seem contradictory to the idea that the initial waves appear due to Kelvin–Helmholtz instability. Indeed, the stability analysis yields a broad range of wave numbers, k , for which the increments of magnitude growth, α , are positive (i.e., such waves are unstable). For a typical shape of the $\alpha(k)$ -dependence see, e.g., Alekseenko et al. (2009). The range of unstable waves starts from $k = 0$ and extends to the “neutral” wave number k_0 ; k_0 is several times larger than the wave number, corresponding to the maximum growth of the waves. On the other hand, the waves can be observable in the experiments only after they have reached some finite amplitude, due to exponential growth over certain distance with the increment $\alpha(k)$. In process of such growth the difference between the amplitudes of the waves with different α will be massively enhanced. We assume that the range of frequencies encompassed by the main ‘Gaussian’ hump (Fig. 12b) corresponds to the close vicinity of the maximum in $\alpha(k)$ dependence. Within this range, α is nearly constant; the waves outside of this range would grow much slower and after passing some distance they will become undistinguishable in comparison to the waves of maximum growth.

The degree of regularity is illustrated in Fig. 12(c, d), in which temporal records of film thickness at a fixed distance ($x_0 = 1.5$ mm) from the inlet are compared to their “linear” reconstruction, $h(t) = A_0 + A_1 \sin(2\pi f_1 t)$. Here A_0 is the first amplitude in the spectrum, and A_1 is the integral of the first term in the

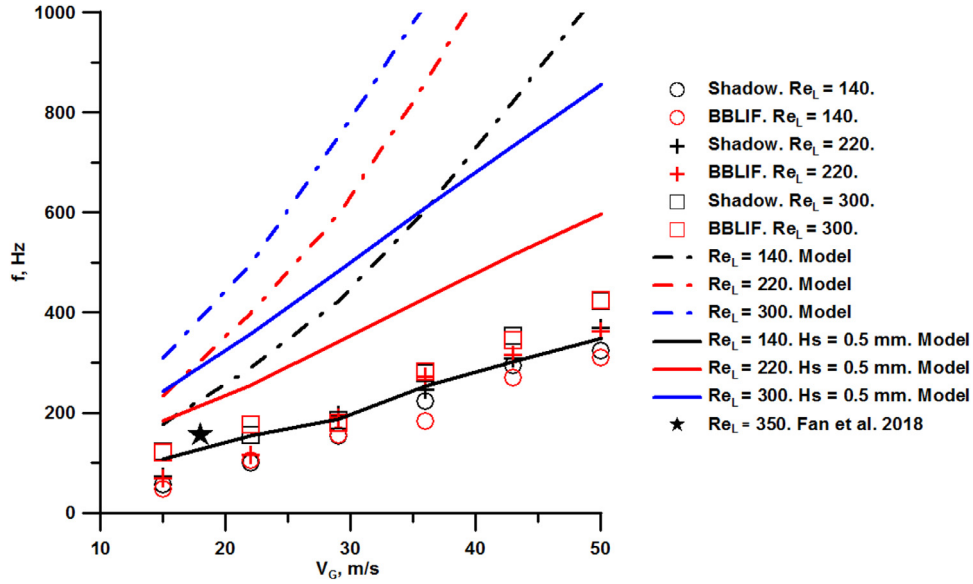


Fig. 13. Passing frequency of the initial waves.

approximation (68). For both the shadow data (Fig. 12c) and BBLIF data (Fig. 12d) the sinusoidal approximation works reasonably well: both frequency and amplitude of the waves can be measured this way. The rough shape of the shadow record is related to limited spatial resolution in the radial direction (about $30\mu\text{m}/\text{pixel}$), which is an inherent problem of the shadow method. The BBLIF record shows much more “smooth” shape; though, at the crests of the initial waves sharp peaks can be occasionally observed. These peaks are caused by the optical distortions affecting the BBLIF data. The maximum slopes of the initial waves at this stage are relatively low ($\sim 10^\circ$ for the considered case). Hence, the distortion is not related to total internal reflection of light at the interface and may rather be attributed to focussing of the reflected light by the curved interface under the wave peak. The estimations show the focus point is located outside the liquid film, but not very far from the wall, given the relatively large average film thickness. Thus, the shadow method is considered a more reliable tool to measure the amplitudes of the waves, as could be expected.

The quasi-sinusoidal character of the waves enables us to facilitate comparison to theories, by limiting the measurements by the so-called area of linear growth of the waves, where simplified linear stability theory is applicable. The solution of a linearised system of equations is sought in form of exponentially growing sinusoidal waves, described as $\sim \exp(ikx - i\omega t + \alpha x)$, where k is the wave number, c is the phase velocity, and α is the increment of spatial growth of the amplitude (note that $\alpha = -k_{im}$ in terms of Section 3.3). The same quantities can be extracted from the experimental records of film thickness; namely, the wave frequency, $f = \omega/2\pi$, the spatial increment of the growth of the amplitude, α , and the phase velocity, c , can be determined.

4.4. Passing frequency of the initial waves

The frequency of the initial waves was measured within the region of flow stabilisation (its length is given in Fig. 11a). The frequency was defined as the position of the main hump in the signal spectrum (see Section 4.3). The error of frequency measurements was estimated as 10% based on ratio of the hump's width to the position of its centre, σ_1/f_1 . Frequency of the initial waves increases with increasing gas velocity and liquid flow rate (Fig. 13, symbols). Frequencies obtained by the BBLIF method and the shadow method agree well with each other.

In order to compare the experimental results to the linear stability analysis, an assumption is made that the initial waves appear from the most unstable waves (waves of maximum growth), i.e., the perturbations with wave number, k , corresponding to the maximum increment of the amplitude growth, α . In Fig. 13 the dashed lines indicate the frequency of the waves of maximum growth, calculated as described in Section 3. The calculated frequencies may reach the values of the order of 1600 Hz at the highest gas velocities, but for clarity of comparison with the experimental data, only the values up to 1000 Hz are shown. Both measured and calculated frequencies grow with gas and liquid flow rates, but the calculated data significantly overestimate the measured values.

It should be noted that this calculation is made for the waves developing on an unperturbed liquid film with equilibrium thickness, H , which roughly corresponds to the “stabilised” value of average film thickness far from the inlet. In the experiments, the waves develop on the thicker layer of liquid prior to stabilisation of the average film thickness. Thickness of the unperturbed layer here is close to the slot thickness, $H_s = 0.5\text{ mm}$, which is several times larger than H . The higher film thickness implies lower liquid velocity in the film, which is expected to affect velocity and frequency of the waves.

Proper modelling of the instability of the non-equilibrium film requires a different velocity profile in the unperturbed liquid film, and, hence, significant improvement of the model. There is a way to roughly estimate the effect of the non-equilibrium film thickness on the waves characteristics, based on re-scaling of the “equilibrium-film” waves characteristics to the new (real) values of average film thickness and average liquid velocity. Assuming that the dimensionless parameters, $\tilde{k}$, $\tilde{c}$ and $\tilde{\alpha}$ (see Section 3), calculated for the case of $h = H$, will be the same for $h = H_s$, and taking into account conservation of liquid flow rate, $q = UH = U_s H_s$, the dimensional quantities, k' , c' , α' , in the non-equilibrium conditions, can be related to the corresponding values k , c and α for equilibrium conditions as follows:

$$\begin{aligned} k' &= \frac{\tilde{k}}{L} = k; & c' &= \tilde{c} U_s = \tilde{c} U \frac{H}{H_s} = c \frac{H}{H_s}; \\ \alpha' &= \frac{\tilde{\alpha} U_s \tilde{k} L}{\tilde{c} U_s + \tilde{k} L \frac{dc_r}{dk} \frac{U_s}{L}} = \frac{\tilde{\alpha} \tilde{k} L}{\tilde{c}_r + \tilde{k} \frac{dc_r}{dk}} = \alpha \end{aligned} \quad (69)$$

Thus, renormalisation will affect the phase velocity and, as a consequence, the frequency of the waves. The modified fre-

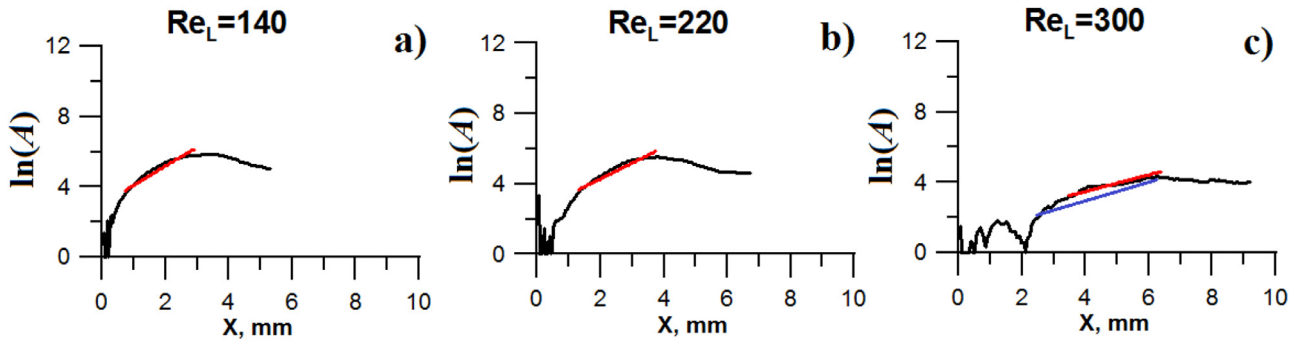


Fig. 14. Change in the spectral amplitude of waves with distance, $V_G = 29$ m/s.

quency values ($f' = k' c' / 2\pi$, Fig. 13, solid lines) show much better quantitative agreement to the experimentally measured frequencies: for the lowest $Re_L = 140$ and $V_G \geq 29$ m/s, the ratio of modelled to measured frequency does not exceed 20%. For the other conditions, the model overestimates the measured frequency approximately twice.

CFD-results obtained by Fan et al. (2018) were also compared to the experiments, though only one combination of flow rates ($Re_L = 350$, $V_G = 18$ m/s) is available at present. The first dominant frequency in the spectrum of the modelled film thickness record is equal 155 Hz, which corresponds very well to the experimental data (it should be noted that the experiments deal with the lower Re_L values).

4.5. Increment of spatial growth of amplitude of the initial waves

To measure the spatial increment of the growth of the amplitude, α , within the spectral approach, it is sufficient to study the dependence $A(x)$, where A is the integral of the first term in the Gaussian approximation of the main hump of a spectrum (see Section 4.3). Examples of such arrays plotted in a semi-logarithmic scale are shown in Fig. 14. The area of the main hump's amplitude growth is restricted by the values x_0 and x_{end} . Upstream x_0 the values of $\ln(A)$ show significant noise, which suggests absence of well-pronounced waves in this area. The threshold value of amplitude to define x_0 was set to $\ln(A) = 2$. x_{end} was determined as the point in which the maximum value of A is reached. Rough estimation of the increment can be determined as the slope of the linear approximation of the dependence $\ln(A(x))$ in the range from x_0 to x_{end} , shown by the blue line in Fig. 14(c).

As can be seen from Fig. 14c, the form of the dependence $\ln(A(x))$ in the region from x_0 to x_{end} is not perfectly straight. Therefore, an alternative method to more accurately determine the increment of growth of the amplitude was proposed: the area of linear growth was determined by finding the longest subarea yielding approximation in form $\ln(A(x)) = px + q$, for which the mean absolute deviation doesn't exceed 5%. These linear approximations are shown by the red lines in Fig. 14. The difference between the data obtained by the two methods (i.e., the slopes of red and blue lines) is about 25%. This value gives an estimate of the measuring error of the growth rate of the amplitude. Further, the data obtained by the second method will be presented.

The spatial growth rate of the amplitude can also be determined using a statistical approach, assuming that ratio of standard deviations of film thickness records obtained at different x is equal to the ratio of real amplitudes of the waves. For each value of x the standard deviation of the signal was measured over the entire temporal length of a record. Further, the dependencies of standard deviation on x were processed in a similar manner as the spectral

amplitude dependencies to estimate the spatial growth rate of the amplitude.

Fig. 15 shows the increment values measured by the spectral approach (thin solid lines correspond to linear approximation of the data) and by the statistical approach (the linear approximation is shown by thin dashed lines). The increment increases with increasing gas velocity and decreases with increasing liquid flow rate. The calculated increment values are shown in Fig. 15 by the coloured lines. They show the same qualitative behaviour with gas and liquid flow rates as the experimental values do. However, the calculations underestimate the values of increment 2–6 times; the difference increases with gas velocity. Alekseenko et al. (2009) did also observe underestimation of the increments by a similar model for a large value of Re_L (equal 125) even at low $V_G = 7.3$ m/s (see Fig. 9c in that work). Estimation of the increment based on the calculations of Fan et al. (2018) yields good agreement to the experimental data, though for this particular combination of flow rates the increment is also described well by the linear stability theory.

Fig. 16 illustrates how the location of the region of exponential growth of the amplitude changes with increasing gas velocity, and confirms the assumption that the initial two-dimensional waves exist in the region of film thickness stabilisation.

4.6. The velocity of the initial waves

One method to measure the phase velocity of the initial waves is based on the spatial gradient of the phase of the main harmonic of the signal, $\varphi = k(x - ct)$. The value of the phase, φ_i , for each value of x_i (Fig. 17, top row) was determined using the spectral method within the linear growth region. After that, the gradient of the dependencies (taking into account the phase shifts by π) was substituted into the relation $c = 2\pi f / (d\varphi/dx)$. The total measuring error for the phase velocity will be the sum of errors of measurements the phase gradient (which does not exceed 3%) and the frequency of initial waves (which is about 10%).

The velocity of the initial waves can be also determined by cross-correlation analysis of the temporal records of film thickness. Thus, the velocity at point x_i will be equal $V_{corr} = 2dx/\tau$, where $2dx$ is the distance between the camera pixels gaining the compared records (dx was taken to be 0.43 mm), τ is the time delay corresponding to the maximum of the cross-correlation function. To increase the measurements accuracy, the vicinity of the maximum of the cross-correlation function was interpolated by a parabola, and the position of its maximum was used as a more precise estimation of τ . The velocity values obtained for all x_i inside the area of linear growth (taking into account the finite dx value) can be averaged to obtain the velocity estimation inside this area (the total RMS-error doesn't exceed 12%).

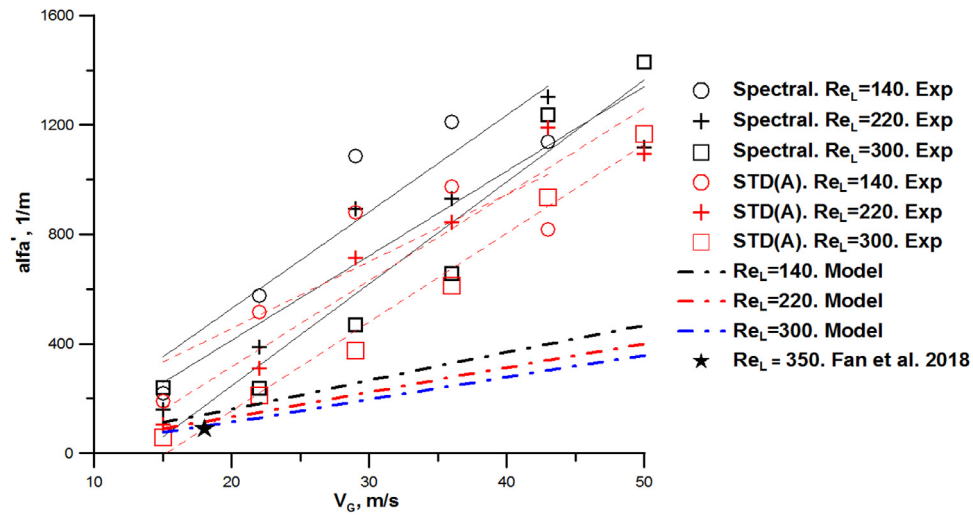


Fig. 15. Increment of spatial growth of amplitude of the initial waves.

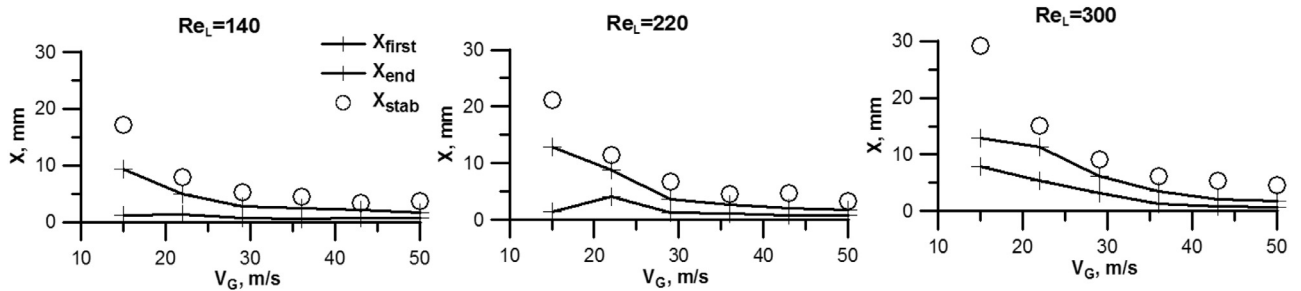
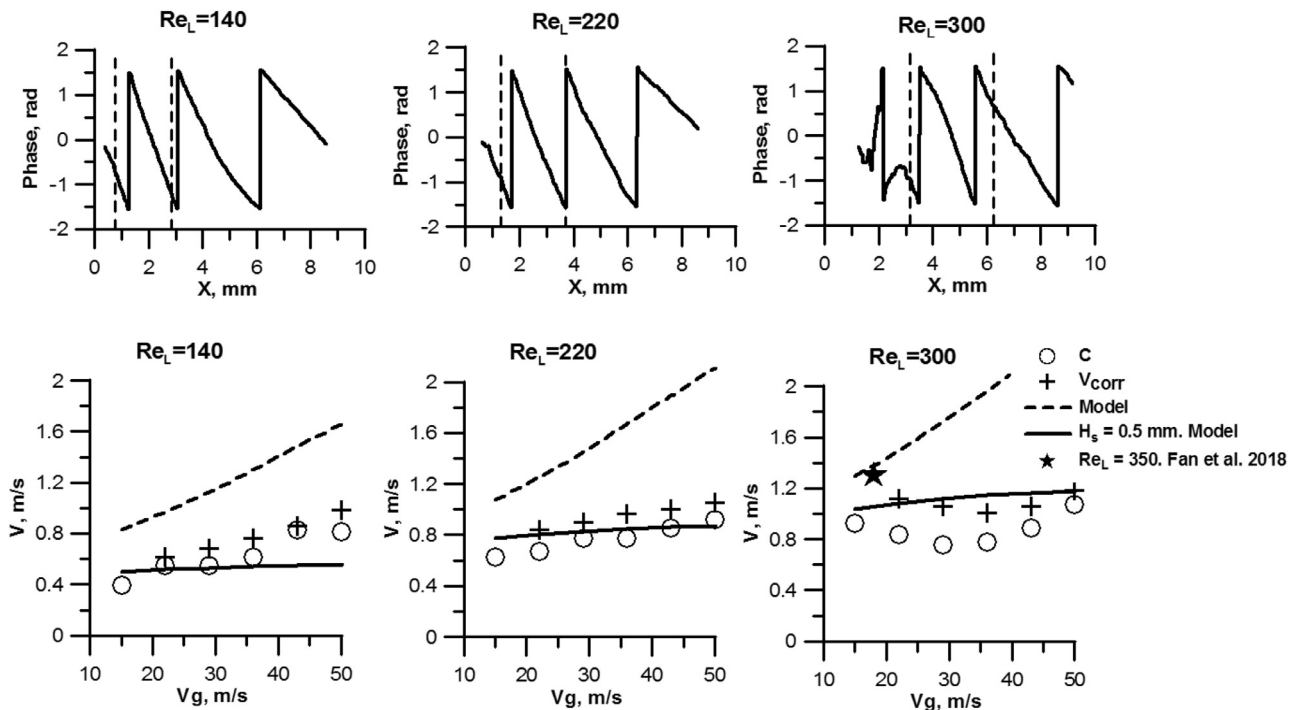


Fig. 16. Area of linear growth of amplitude (between the crosses) and thick layer of film (circles).

Fig. 17. (top) Change of the phase with distance. $V_g = 29$ m/s; (bottom) Velocity of the initial waves.

It can be seen from Fig. 17 (bottom row) that the velocity of the initial waves increases with the gas velocity at $Re_L = 140$ and 220 and demonstrates more complex behaviour at $Re_L = 300$. The values of the phase velocity (c) show reasonable agreement with the values of the cross-correlation speed (V_{corr}).

The calculated velocity values for the equilibrium film thickness (dashed lines in Fig. 17, bottom row) show strong increase with the gas velocity and significantly exceed the experimental data (only the values below 2 m/s are shown). The modified calculations (solid lines) show more gradual increase of waves velocity with the gas velocity and demonstrate much better agreement with the experiment. The CFD model by Fan et al. (2018) shows slight overestimation of the phase velocity of the initial waves.

5. Conclusions

The behaviour of the liquid film at the initial stage of the dispersed-annular flow was studied using the brightness-based laser-induced fluorescence (BBLIF) method and the shadow method. The experiments were carried out in downwards adiabatic air-water flow in cylindrical and rectangular ducts in a wide range of liquid and gas velocities.

Using the BBLIF method, the process of formation of disturbance waves was studied in three-dimensional approach; it consists of several stages. The spatial length of each stage decreases with increasing gas velocity, but increases with liquid flow rate. The length of the region of existence of the initial two-dimensional waves is practically independent on the shape of the channel.

Using the two-dimensional approach of the BBLIF method and the shadow method, the evolution of liquid film in the vicinity of the inlet was investigated. It is established that the initial two-dimensional waves first appear at the thick initial film prior to stabilisation of film thickness. The initial waves were found to be highly regular and were analysed within the line of thought typical to the linear stability analysis, by measuring the frequency, phase velocity and increment of amplitude growth of the waves. All three quantities increase with the gas flow rate and undergo less pronounced dependence on liquid flow rate. The area of the linear growth is also limited to the thick initial film and the lengths of the areas show the same behaviour with gas and liquid flow rates.

The experimentally obtained waves characteristics near the entrance were compared with the results of linear stability analysis. Stability theory overestimates several times the frequency and phase velocity of the initial waves; the discrepancy increases with gas and liquid flow rates. Better agreement can be achieved using renormalisation based on local non-equilibrium film thickness and, hence, liquid velocity. In such a case the discrepancy is reduced to 10–20% for the frequency of the initial waves for the lowest Re_L ; the predicted frequency remains about 2 times larger than the measured values for larger Re_L . For the phase velocity, the discrepancy is also massively reduced by means of renormalisation. The calculated values of the increment of magnitude growth are underestimated 2–6 times compared to the experimental ones. Still, general agreement in frequency and velocity indicates that the initial waves can be described within the frames of Kelvin-Helmholtz instability analysis. Further development of theoretical models applicable to the case of large gas and liquid flow rates together with appropriate methods of taking into account the initial stage of the flow development is required. Afterwards, the nonlinear development of the initial waves and their transverse instability leading to break-up into three-dimensional waves should be modelled.

Conflict of interests

None declared

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ijmultiphaseflow.2018.11.010.

Appendix. Measurement of the film thickness by the shadow method

To measure the local thickness of a liquid film on the basis of images obtained by the shadow method, it is necessary to provide justification of the correspondence between the image size and real film thickness. To do this, we consider the optical problem of the path of the rays from the light source to the camera (see Fig. 18), to accurately indicate the position of the inner wall of the channel and the gas-liquid interface in the image. Table 1 shows the main parameters of the regarded problem (see also Fig. 2(a,b)).

For simplicity it is assumed that the rays leading to the camera are parallel to the x -axis. In such a case, the y -coordinate of an object in the camera image is defined as $R_1 \sin \alpha_1$, where α_1 is the azimuthal coordinate of the point in which a ray that has passed through the pipe reaches its outer surface (see Fig. 18). To simplify the analysis, the case was first considered in which the whole pipe is filled with working liquid, prior to proceeding to the case of liquid film flow. The solution of this problem is to determine the angle at which a ray should enter the pipe to reach the point defined by the particular value of α_1 . Such a ray may travel within the plexiglass pipe only (Fig. 18a) or through the liquid, intersecting the circle of radius R_2 (Fig. 18b). In the first case, the solution requires Snell's law in points 1 and 2 and identifying the coordinate of the intersection between the straight line and the R_1 -circle. For the second case, Snell's law is applied in all 4 points and three equations for the intersection of the ray with the circles R_1 and R_2 are required. It was found that none of the incidence angles reaches the angle of total internal reflection. The system of equations is solved numerically to obtain the coordinates of the entrance point and the angle between the incident ray and the normal to the pipe's outer wall. These data can be used to assess whether the ray belongs to the diffusing sheet.

Fig. 19 shows how the real size of the region along the y -axis (the distance from 0 to the red line) correlates with the image size ($R_1 \sin \alpha_1$). It was found, that the real size will be 1.33 times smaller than the image size in the range from 0 to R_2 and 1.45

Table 1
Parameters of the problem.

Outer pipe radius (R_1), mm	14
Inner pipe radius (R_2), mm	5.85
Slot width (H_s), mm	0.5
Refractive index for air (n_1) / plexiglass (n_2) / water (n_3)	1 / 1.5 / 1.33
Distance from the diffusing sheet to the outer wall, mm	150
Width of the diffusing sheet, mm	200
Angle of incidence of the ray leading to the camera on the outer wall (α_1)	parameter

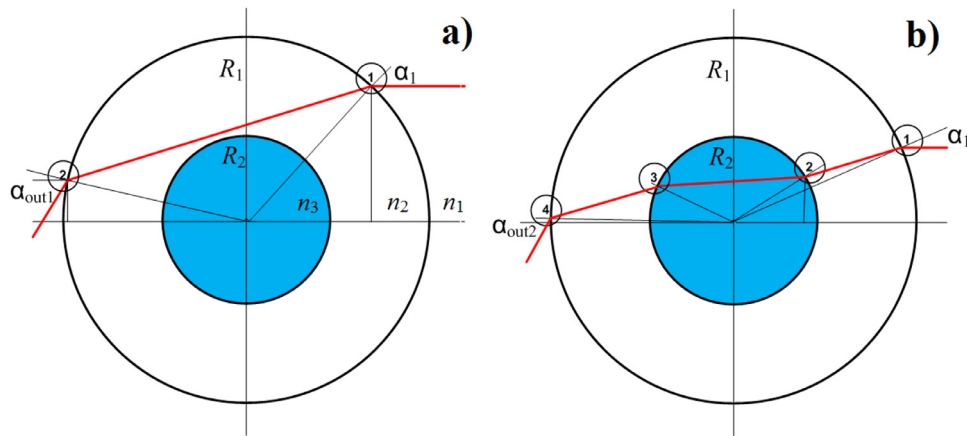


Fig. 18. The path of the rays from the camera (on the right) to the light source (on the left) through a cylindrical channel.

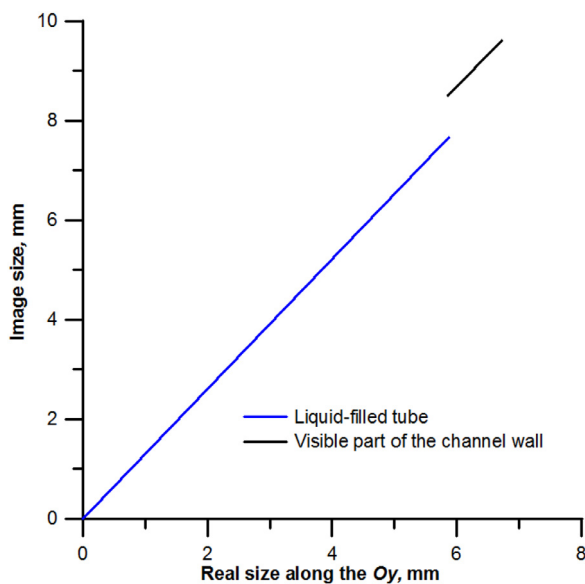


Fig. 19. The ratio of the real size along the y-axis to the image size.

times smaller in the region from R_2 to R_1 . There is also the blind zone near the plexiglass-water boundary, indicating that the rays passing through these points ($R_2 + \delta R_2$) do not reach the camera lens.

Thus, the camera “sees” the entire inner cross-section of the pipe along the y-axis (in the range from 0 to R_2), but this is not enough to obtain the image of the film by the shadow method. It

is also necessary that the entire cross-section under consideration is illuminated by the light source. Let’s consider how the brightness of the image of the section along the y-axis changes. Knowing at what angles the rays enter (α_{out1} and α_{out2}) the outer surface of the pipe, it is not difficult to find at which points they reach the light source (diffusing sheet). Assuming that the light source uniformly illuminates the sheet, correspondence between the length of a small segment of the camera image (ΔY_{cam}) and the width of portion of the light source illuminating it (ΔY_l) can be established as shown in Fig. 20. The higher ΔY_l , the brighter is the image corresponding to ΔY_{cam} .

Taking for the “basic brightness” the size ΔY_{l-0} obtained for the image in the interval from 0 to ΔY_{cam} , we get the dependence of the image brightness on the Y coordinate (Fig. 21). The highest illumination is observed at the water-wall boundary (red circles). Outside this boundary there is a section with zero illumination (see also Fig. 19). After that there is a bright strip corresponding to the visible area of the plexiglass channel wall (green circles), and behind it again the blind zone. The obtained dependence correlates well with the experimental data (shadow image of the falling liquid film is presented in the same Figure). The calculated size of the blind zone near the plexiglass–water boundary coincides with the size of the dark zone in the experimental shadow images. Thus, it can be concluded that the bottom boundary of the film is visible in the image and corresponds to the position of the inner wall of the channel.

The real problem represents the flow of a thin liquid film along the inner surface of the tube. Analysis of the calculated data obtained for the liquid-filled pipe showed that there are rays passing through the points $(R_2 - H_s) + \delta$, where $\delta < H_s$ and do not pass through a circle of radius $R_2 - H_s$ (i.e., do not touch the interface),



Fig. 20. The scheme of the path of the rays from the chamber to the light source through the “flooded” pipe.

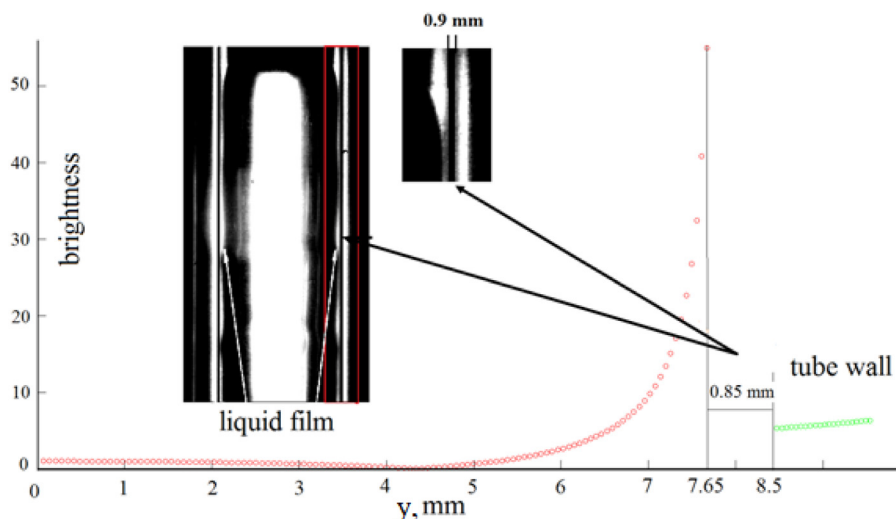


Fig. 21. Modelled and experimental illumination along the y-axis of the camera image.

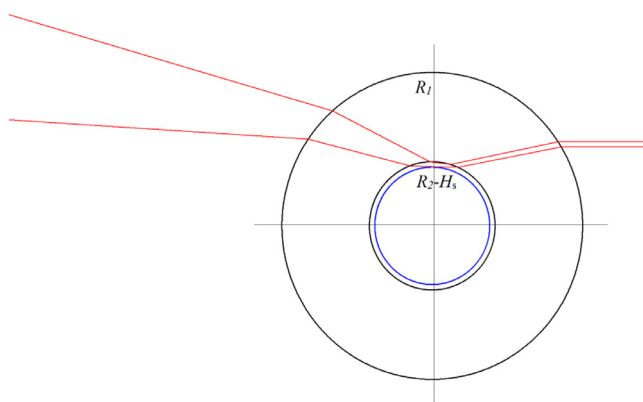


Fig. 22. The optical path of light passing through the liquid film.

as shown in Fig. 22. The angle of inclination of such rays relative to the x-axis is small ($\sim 1\text{--}2^\circ$), which indicates the entire profile of the liquid film is visible.

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